

Wollo University
CMHS
School of Public Health
Department of Health Informatics

Introduction to eHealth for 3rd Year Health Informatics

Students

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Introduction to eHealth system

- After the end of this session students will be able to:
 - Define eHealth
 - Identify and know the components of eHealth
 - Know Ethiopian eHealth policy and strategy
 - Identify the country eHealth experiences

Introduction eHealth

- A relatively recent term for healthcare practice supported by electronic processes and communication, dating back to at least 1999.
- The use of modern ICT to meet needs of citizens, patients, healthcare providers, and policymakers.
- eHealth is the **cost-effective and secure use of ICT** in support of **health and health-related fields**, including **healthcare services**, **health surveillance**, health literature, and health education, knowledge and research.

Cont..

- **eHealth** is an emerging field in the intersection of **medical informatics** and **public health**, referring to **health services and information delivered** or enhanced through the Internet and **related technologies**.
- **In a broader sense**, the term characterized by technical development, **state-of-mind**, a way of thinking, an attitude, and commitment for networked, global thinking to improve healthcare locally, regionally, and worldwide by using information and communication technology.

Cont...

- The application of Internet and other related technologies in the healthcare industry to improve the access, efficiency, effectiveness, and quality of clinical and business processes utilized by healthcare organizations, practitioners, patients, and consumers to improve the health status of patients.

eHealth dimensions

- **Efficiency** -through cost reductions
- **Enhancing** -quality of care
- **Evidence-based** -interventions
- **Empowerment** -of consumers and patients
- **Encouragement** -of new relationships
- **Education** -through online sources
- **Enabling** -information exchange & communication
- **Extending** the scope of health
- **Ethics** in patient-physician interactions
- **Equity** -in the provision of healthcare

Topics in eHealth

❖ **HIMSS** (Healthcare Information and Management Systems Society),

sponsors studies on topics that are current and relevant to eHealth care:

- Telemedicine/ Telehealth ,m-Health
- Electronic health records
- Integration and interoperability
- Patient safety and quality outcomes
- Privacy and security, Clinical decision support and many others

Evolution of eHealth

The term eHealth gradually evolved with the evolution of the applications of ICT in Health

- 1960s ➔ Electronic data processing
- 1970s ➔ Medical Informatics (computing in medicine ➔ Administration and Finance; statistics and Epidemiology; modeling)
- 1980s ➔ **Health Informatics** (Computer-assisted image and voice processing)
- 1985s ➔ Telemedicine (The practice of medical care using audio, visual and data communication) ➔ healthcare delivery, diagnosis, consultation, treatment, education, and transfer of data
- 1990s ➔ Telehealth (e-transcriptions and e-commerce) ➔ ePrescriptions

Evolution of eHealth

➤ 2000s → eHealth

- Clinical care of the individual: **Telemedicine**
- Home health care: **TeleHome care**
- Management of clinical care: e.g. **EHR, ePrescriptions, ...**
- Monitor & control Public Health: **Tele Surveillance**
- Human resources development: **Tele Education**
- Governance of Health Services (includes administrative & financial transactions, and routine services, e.g. those requiring form-filling & submission): **health-related parts of eGovernment, and eTransactions**

Cont....

➤ The next technology boost to eHealth: **Sensor Technology**

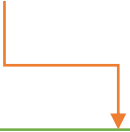
➤ Temperature

➤ Pressure

➤ Texture/feel

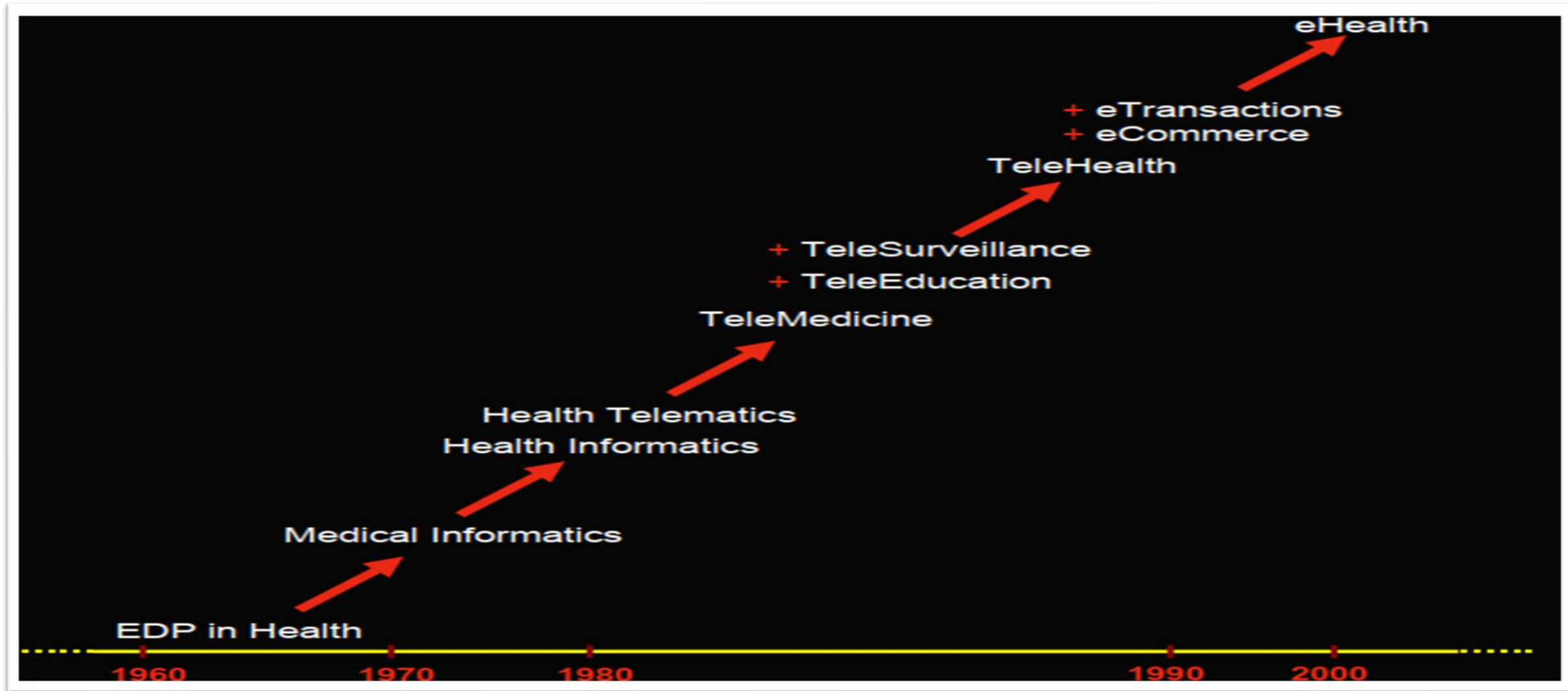
➤ Shape

➤ Smell and etc...



a technology that uses sensors to acquire information by detecting the physical, chemical, or biological property quantities and convert them into readable signal.

Evolution of eHealth....



Components/ Types of eHealth

- **EHR** or **EMR**: systematized collection of **patient** and **population health data** in a digital format that is electronically-stored
- **CDSS**: ICT **solution** designed to provide health professionals with clinical decision support such as assistance with clinical decision-making tasks
- **TM**: physical and psychological diagnosis and treatments at a distance, including **tele-monitoring of patient functions**
- **mHealth**: use of appropriate digital technology for public health ➔ WHO, 2018

E-Health as enabler

- **eHealth** system is an enabler for more **integration**, **improved access**, and better health care through better information management and use.
 - **Information exchange between providers** on **secure networks** to enable providers to have timely access to critical information at the point of care;
 - Telemedicine help to address access to health services;
 - Electronic health records , **consolidated health information**, and **integrated decision support tools** at point of care to reduce medical errors, **improve health system efficiency**;
 - Internet enabled consumer health information tools to empower consumers to interact with care providers in new ways and to manage their health more actively.

eHealth Drivers

- Technology availability and capability
- Consumer and practitioner uptake
- Health system demands
 - Social demography
 - Health costs
 - Equity and access
 - Geography
- Innovation

eHealth Interventions

- Delivering health information to health professionals and **health consumers through the Internet and telecommunication**
- Using IT and e-commerce to improve public health services for example, through education and training of health workers
- Applying e-commerce and e-business practices in health systems management (planning, budgeting, accounting, etc.)

Four Layers of Intervention

- Application layer
- Operation layer
- Infrastructure layer
- Transmission layer

Application of eHealth for Health management information system

- Data analysis, presentation, and use

- Extraction and triangulation of health data

- Data analysis and synthesis tools to produce routine or ad hoc reports, identify trends and issues of concerns, and track progress

- Data dashboards and other data visualization tools can be customized to include maps (geographic information systems, or GIS)

- Data quality

- Tools for improving data quality include data standardization, geocoding, data matching, data monitoring, and profiling.

Benefits of eHealth

➤ Health information

- Evidence / information by mode for consumers/practitioners
- Systems for communication over distance and over time
- Clinical decision supports???

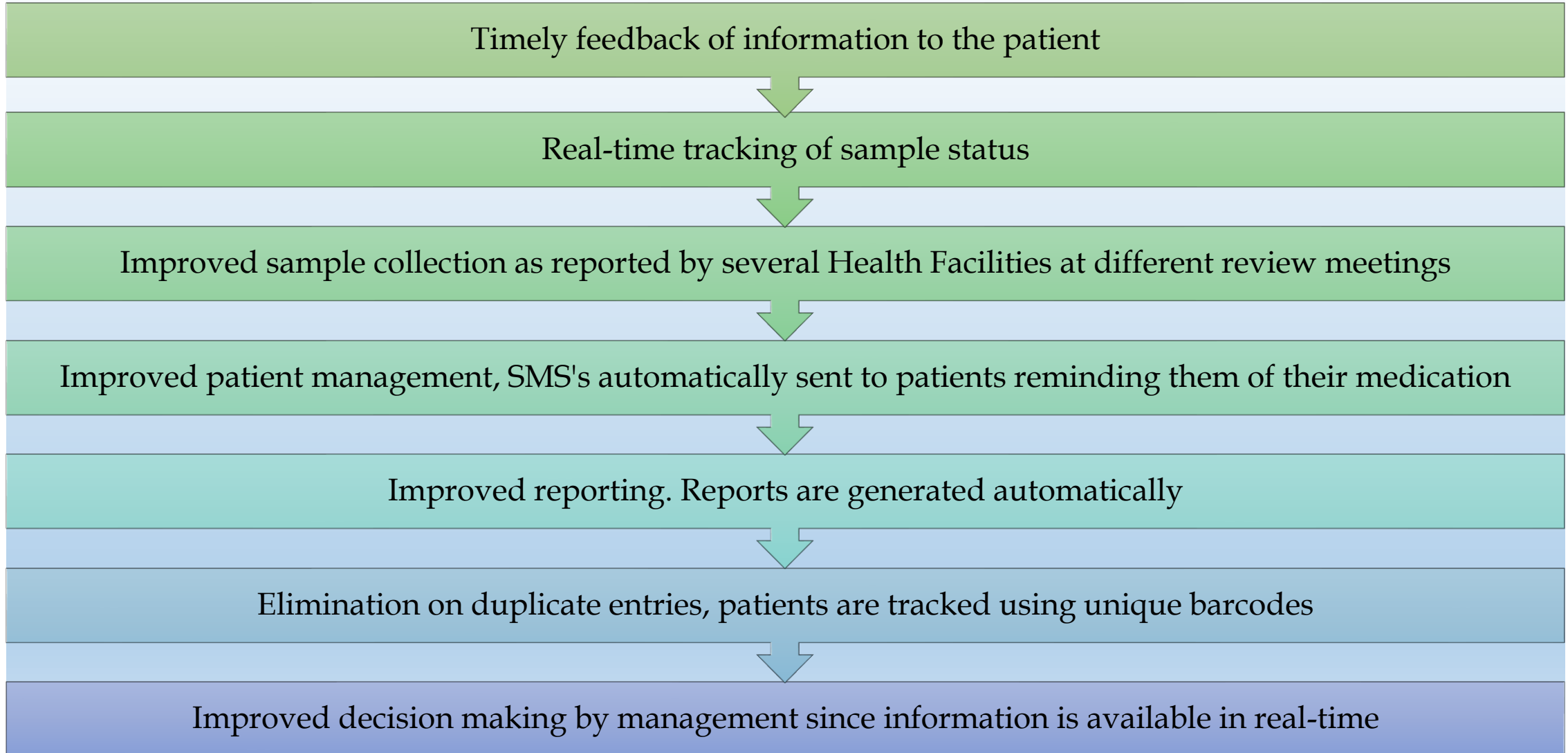
➤ Improve public health

- Education and training (formal; professional development)
- Data mining, Surveillance

➤ e-commerce / e-business

- eHealth record, ePrescriptions
- Appointment scheduling

Benefits of eHealth...



Major Challenges of eHealth Systems

- **Security and privacy**
- **Confidentiality:** assurance of non-disclosure of sensitive data
 - confidentiality attacks: eavesdropping, location tracking (mHealth), and activity tracking (eHealth monitoring systems)
 - Eavesdropping is when an attacker attempts to access medical data while it is acquired, transmitted, or stored either remotely or locally
 - Confidentiality attacks not only lead to a breach of patient's privacy but also potentially allow adversaries to modify certain data resulting in wrong diagnosis or treatment

....

- **Integrity:** preventing unauthorized data modification
 - An eHealth system should contain a tool allowing to detect any changes in patient medical documentation
- **Availability:** service or data should always be available when it is needed
- **Anonymous:** Anonymity to Third Parties

Challenges of eHealth implementation in developing countries

- **Lack of data and Communication Technology (ICT) Infrastructure:** Computers, communication devices (smartphone, iPad...)
- **Computing instrumentation**
 - Computing instrumentation (desktops, laptops, servers),
 - Transmission systems (television sets, VCD and optical disk players, camera's and camcorders),
 - Networking devices (switches, routers, wireless access points, firewalls, native space networks (lans) and wide space networks (wans)),

- Imaging (desktop and network printers)
- Mobile telephone and communication (personal digital assistants (pdas), cell phone, landlines, fax machines etc.)
- Net systems (modems)
- **Initial stages** : ICT infrastructures that are presently in place are not adequate to support the new system
- **Electric Power Supply**
- **Basic ICT Knowledge and Skills:** level of ICT knowledge and skills amongst both healthcare providers and consumers is so little

➤ **Waste of time:** there is a view that healthcare professionals who have deficiency in ICT skills of management the online health data end up using too much time

➤ **Training of human**

➤ **Skilled workforce**

➤ Skilled workforce that understands health care, information, communication technology, people and organizational structures

➤ One of the most frequent challenges among eHealth projects in Africa

➤ **Resistance to New Technology**

- **Perceptions:** Users' perceptions about technology can affect exactly how a new information technology is conceptualized, will be accepted and further used in the community in which it will be operationalized
- **Internet:** The rate of low Internet penetration and low bandwidth are some of the challenges to eHealth implementation in developing countries

- **Financial and Sustainability Issues:** To some stakeholders, implementation of new eHealth system means the health care professionals and the rural folks in the villages must own some high-speed laptop or desktop computer, smartphones (iPhone, Android), or iPad, subscribe for Internet service and remunerate monthly subscription fee
- **Funding:** studies show positive relationship between ICT implementation and organization size since there is more funds in the big hospitals compared to smaller institutions like clinics

Challenges of eHealth implementation in developing countries

- **Sustainability:** corrective and preventive maintenance is extremely important for any ICT instrumentation
- **Data protection :** strategies and processes to secure the privacy, availability and integrity of data
- **Regulatory of legal and policy framework:** Regulatory of legal and policy framework is tough in most developing countries to get clear policies and coordination between governmental agencies and eHealth initiatives
- **Socioeconomic Constraints and Development**

Challenges of eHealth implementation in developing countries

➤ Unavailability:

➤ Unreliability:

➤ Inaccessibility:

➤ Lack of skills:

Challenges of eHealth in Ethiopia

- eFoundations (standards, interoperability, security and access)
- National infrastructure including **health identifiers** and clinical terminology
- Implementation activities
- Knowledge
- Integration and coordination
- Policy development (eHealth strategy/master plan)
- Capacity and skills
- Lacking of hard evidence and **impact evaluation**

Ethiopian eHealth policy and strategy

eHealth: serving a higher purpose



- The Health System Molecule
- *Putting people at the centre of the health system*
 - **Stewards**
 - **Providers**
 - **Patients**
- *And eHealth underpinning health systems*

eHealth: people, processes and technology

- eHealth revolution predicated on trinity of people, processes and technology
- People are the key, because although technology and processes can change people, the greater influence pathways are those of people changing processes and technology - through inventiveness innovation, creativity
- Organize the people to harness the power of eHealth

“eHealth” a broad and diverse realm of efforts

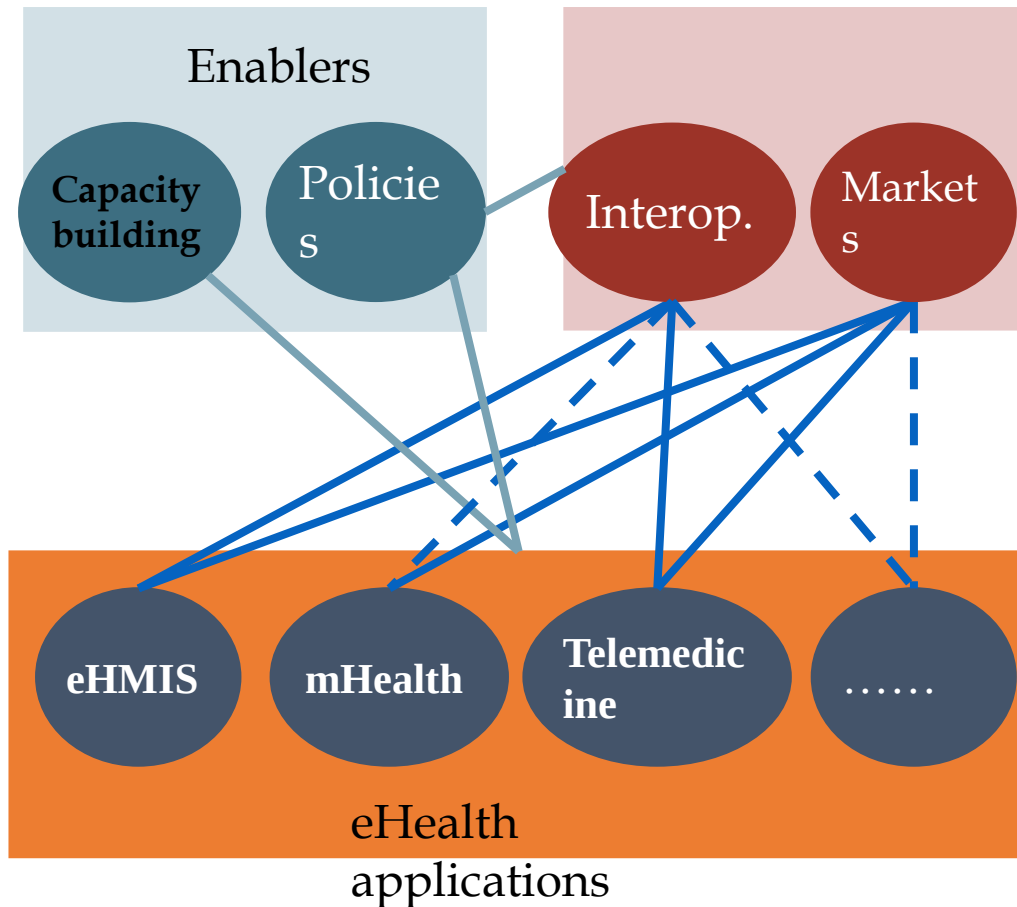
eHealth: *the use ICT to improve health*

Different types of eHealth initiatives include, but not limited to:

- Health information systems
- *Public health informatics:*
 - Support for disease prevention
 - Disease and intervention surveillance (e.g. PDAs to community health workers for disease surveillance)
 - National health info systems to detect/track global threats to public health
- ***Health and clinical informatics:***
 - Electronic health records (EHR), electronic medical records (EMR), patient health records (PHR)
 - **Decision support for healthcare professionals**
- *Health system administration and operations*
 - Pharmacy and supply chain management systems
 - Laboratory systems (e.g. electronic ordering, transmission processing)
 - Clinical administration software (e.g. billing)
- Healthcare and expertise
 - Telemedicine / telehealth
- ***Health research, advisories and education***
 - eLearning for physician, nurse, healthcare personnel training
 - Access to research for healthcare personnel
 - Patient support and information (SMS reminders for drug compliance, online health information, etc.)
 - Decision support for healthcare professionals

eHealth application

Optimal development path



- **“Path to Interoperability”** and “Unlocking the eHealth Market” support developing countries in finding optimal development path to maximize eHealth potential
- “eHealth Policies” and “Capacity Building” address the enabling environment to lower the barriers and impediments to eHealth diffusion and advancement ????
- “Electronic Health Records”, “mHealth”, “Public Health Informatics” and “Access to Information” provide grounding in applications that strengthen health systems

Policy issue 1: Safer decisions, quality of care processes

- Increase the influence on **risk management** (e.g. Medical errors and patient's errors) by timely providing adequate knowledge to assist proper decisions
- Increase the **quality of care processes**, i.e. Declaring and following explicit reference clinical pathways
- Increase the mutual awareness about what other clinicians are **knowing, doing or planning on the patient**

Policy issue 2: Sustainable evolution of healthcare

- The increasing cost of healthcare requires a **rationalisation** of services provided without a negative effect on quality of care **by**
- **Continuity of care,**
- **Patient empowerment,**
- **Accurate governance** based on routine data (with **timely indicators**, also to allow for self assessment of healthcare professionals)

Policy issue 3: Improve access to services

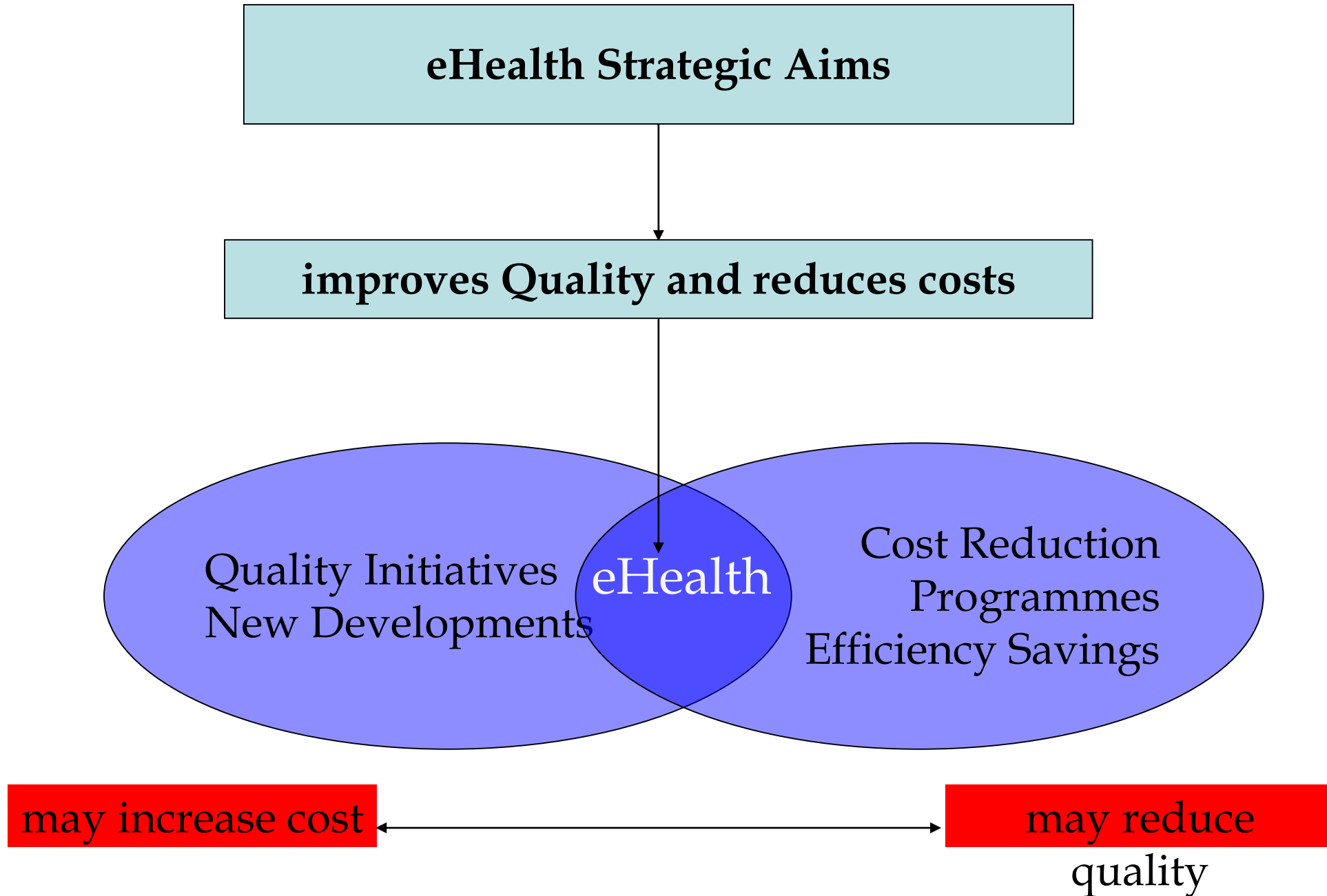
- Simplification of the paper work
- Rationalisation of organisational and administrative processes
 - Increase of efficiency of operational workflows (e.g. **prescriptions booking reports**).
- Effective portals, with practical information and authoritative clinical knowledge

MOH vision for eHealth

- Ambition **to use IT** to transform services while retaining an incremental and pragmatic approach
- strategic eHealth aims
- Has the citizen at the centre
- eHealth is central to improvements in quality and can bring about efficiencies
- An agreed direction and set of goals

MOH Intention

- Build on **direction of travel** and **achievements**
- Recognise the importance of clinical leadership and engagement, and incremental development
- Move from a focus on technology towards a focus on benefits and outcomes
- Promote, encourage and facilitate collaboration between Boards
- Encourage convergence (**EHR**)
- Be pragmatic and incremental



Aim

- The National eHealth strategy of Ethiopia is aimed at guiding and streamlining the Information and Communication Technology (ICT) solutions in the healthcare sector.
- The core of the strategy is the commitment that health program will be focused on three groups of beneficiaries
 1. health and healthcare providers,
 2. Consumers of health and
 3. Health and healthcare service managers.

Vision and Mission

- **Vision:** Provision of equitable , quality, and timely health services through the use of e-health.
- **Mission:** To adopt ICT solutions appropriately in the healthcare sector of Ethiopia, to improve the health outcome by bridge the equity gap and enhancing the effectiveness and efficiency of the health services delivery.

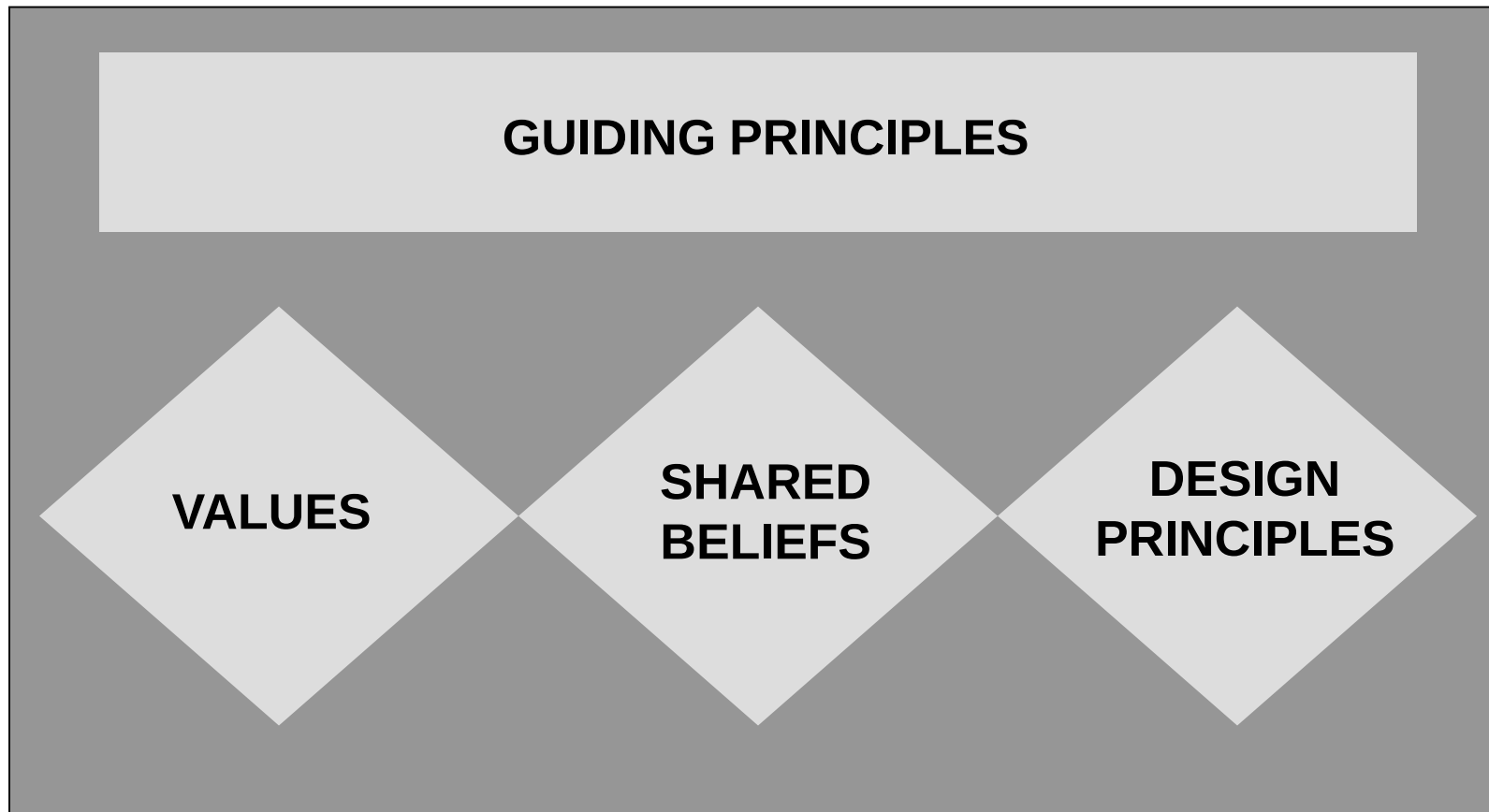
Objectives

- The main objective of eHealth strategy is to utilize ICT to generate, capture, transmit, store and retrieve digital data for clinical, educational and administrative purposes.
- **Specific objectives**
 - To improve access, quality, efficiency of the health care services
 - To enhance decision support system through a timely, accurately and comprehensively data/information management system
 - To improve the referral system and reduce duplication of service delivery
 - To continue the human resources development
 - To support the health research operations
 - To support public information dissemination on health service
 - To provide standards for ensuring interoperability on the health sector

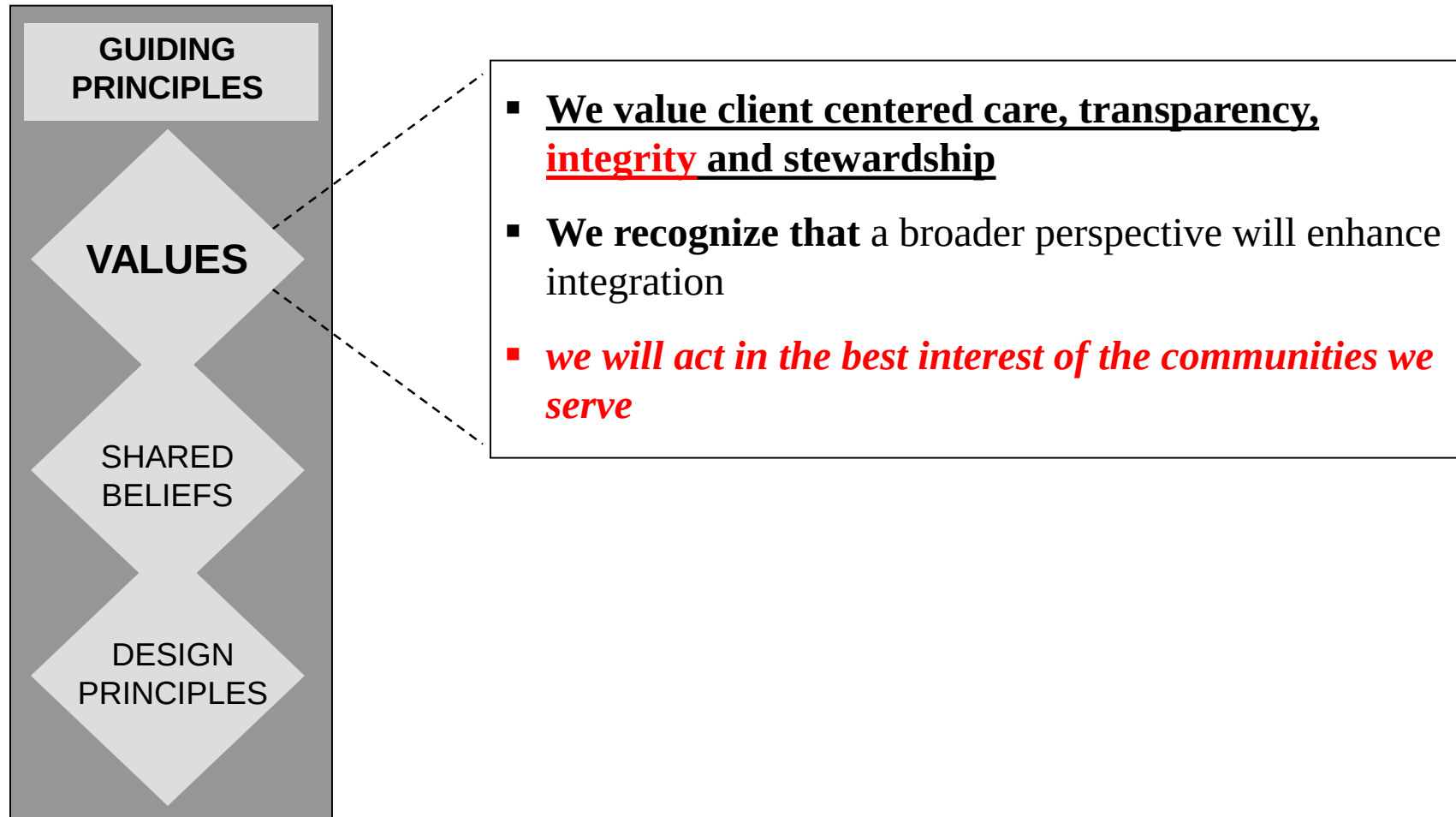
Strategic areas of interventions

- Implementing the national e-health infrastructure and rules to allow information to be seamlessly accessed and shared across the Ethiopia health system.
- Developing management information systems and applications that can deliver tangible benefits to public, care providers and health care workers.
- Establishing governance and leadership regime to enable effective coordination and oversight of eHealth activities.
- Educational promotion to all stakeholders on technology usage

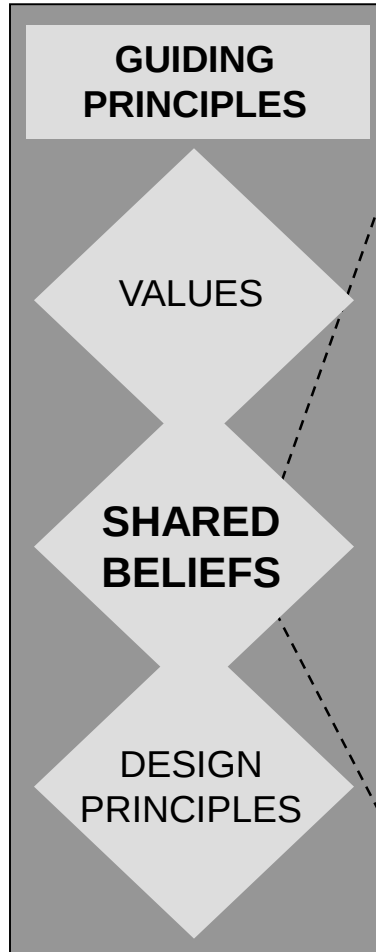
Guiding principles will lead the development of the e-Health strategy



Values are a component of the guiding principles that will lead the eHealth strategy

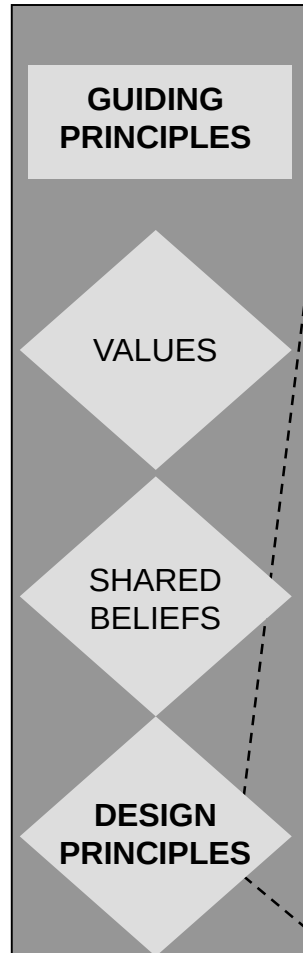


Shared beliefs



- e-Health is a critical enabler to:
 - Achieving integration of service delivery within and across the country
 - Making it easier for people to do their jobs well
 - Engaging people in their own health and wellness
 - Bringing sustainable value to the health care system
 - Ensuring better health outcomes
- Through collaboration, there are things one can achieve in e-Health that one cannot achieve by acting alone
- Performance builds confidence by demonstrating success early and continuously, then momentum will build attracting financial support from outside sources
- **Primary care is at the heart of health service integration and management**

Design principles are a component of the guiding principles that will lead the e-Health strategy



- Design e-Health initiatives around the client, which will also meet the needs of providers and organizations
- It is important that all e-Health initiatives have clearly defined and measurable health, clinical, and/ or business outcomes
- It is important that the e-Health strategy reflects both in terms of needs and capabilities
- Recognize there is a broad range of IT-readiness in our community, and acknowledge that:
 - stakeholders must experience progress in e-Health
 - “One size will not fit all”
- Have a balance of **“quick wins”** with initiatives that will demonstrate value over a longer term
- Recognize that success depends on maintaining focus and approaching large complex initiatives in manageable increments; ***We will not try to “boil the ocean”***
- Leverage existing investments and capabilities; and not **“reinvent the wheel”**
- Respect people’s individual needs for privacy and confidentiality, and ensure compliance with all legislated requirements
- Ensure all initiatives are deployed using standards and evidence based practices

Summary of Guiding Principles

- Phase implementation of e-health initiatives in line with HSDP/HIS/ICT frameworks
- Ensure interoperability and adopt common standards for collaboration and partnerships to share information and services in line with one plan, one budget and one report.
- Consider healthcare service integrity, confidentiality, privacy and security for data/information interchange and management
- Continue to support and starting e-Health programs on available human, financial and technical resources
- Coordinating disparate health and IT experts, recourses (financial and ICT)
- Engage key health care stakeholders in the design and delivery of e-Health solutions.
- Under the guidance and support of NAC, establish a governance mechanism for successful M&E of eHealth programs (eHealth steering committee, HITCT)

Overarching strategic initiatives

1

Provide the infrastructure and enablers required for the achievement of all strategic initiatives noted in the IHSP

2

Improve information sharing within communities and across the continuum of care

3

Improve the management and prevention of chronic diseases

The strategy pillar

- HIS: EMR, eHMIS, pharmacy and medical supply chain Information, management, Financial Information, Human Resource and Laboratory Information System
- Telemedicine: patient centric services, referral system, remote and rural setting
- Mhealth: HDA/HEW/MDG and HSDP targets
- E-learning: Health workforce training and CME
- Community information

Quiz1=10%

1. Define e-Health? (2%)
2. What are sensor technologies and give at least 3 examples. (3%)
3. Explain the barriers and impediments to advance and diffuse eHealth in Ethiopia? (2%)
4. Identify the three groups of beneficiaries that the eHealth strategy of Ethiopia focused on? (3%)

Thanks!

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Chapter-2
Electronic health record

Mulugeta D

Outlines

- Electronic health record (EHR)
- Electronic medical record (EMR)
- Personal health record (PHR)
- Benefit and function of HER/EMR

Electronic Health Records as a Part of Health Information Systems

- The most important enterprise functions in health care are related to diagnostics and therapy.
- Obviously, data that are relevant to medical decision making need to be collected and presented in a patient record
- A patient record in general is composed of all data and documents generated or received during the care of a patient at a health care institution.

Electronic health record

- Nowadays, many documents in the paper-based patient record are computer printouts, such as laboratory results, or discharge summaries typed into a text processing system
- The portion of documents created in computer-based form will further increase
- It seems natural to strive for a patient record that is partly or completely stored on electronic document carriers: the *electronic health record (EHR)*

EHR...

- EHRs are needed to support **functions of patient care** as well as for **administrative functions**
- The collection of medical data relating to one subject of care, i.e., the patient, that is stored in the computer-based part of a health information system
- **“The EHR for a subject of care might be scattered physically across multiple (discrete or interconnected) clinical systems and repositories, each of which will hold and manage a partial EHR for each of its data subjects, scoped according to the service or community settings, clinical domains and time periods of use of that system in the life of each person.”**

EHR...

- EHR used to be **provider-centric**, i.e. they only contained patient information that was recorded in one institution, e.g. in a hospital or in a physician's office. Those EHR are usually called **electronic patient records (EPR)**
- Hence, potentially relevant information about the medical history of a *patient* that was recorded in other institutions was missing or had to be recorded again. This led to **quality and efficiency** problems.

EHR...

- Although this situation can still be found in many institutions, efforts are being made today to organize EHRs patient-centric, i.e. independent of institutional boundaries.
- To achieve the vision of a complete and lifetime-spanning EHR which supports health care on the one hand, but respects legal and ethical issues on the other hand, different strategies can be found.

Functions of EHR

- **Primary**: support patient care by providing relevant information about a patient whenever and wherever it is needed.
- **Secondary**: administrative functions, such as billing and quality management

❑ Types of EHR

1. ***Provider-centric EHR***: Only contained patient information that was recorded in one institution, example in a hospital or in a physician's office called **Electronic patient record(EPR)**
 - ✓ Hence, potentially relevant information about the medical history of a patient that was recorded in other institutions was missing or had to be recorded again. This led to quality and efficiency problems.
2. ***Patient-centric EHR***: Independent of institutional boundaries

Electronic Health Record

- Real-time longitudinal electronic record of an individual patient's health information that can assist health professionals with decision-making and treatment.
- Patient demographics, progress notes, problems, medications, vital signs, past medical history, immunizations, laboratory data, radiology reports.
- EHR can automate and streamline clinician's workflow.
- Can generate a complete record of a clinical patient encounter, as well as supporting other care-related activities directly or indirectly via interface - including evidence-based decision support, quality management, and outcomes reporting.

Electronic Health Record (EHR)

- Electronic record of health-related information on an individual **across more than one health care organization.**
- Used by a variety of healthcare providers: A record that is available electronically to authorized health care providers and to the individual patient anywhere and anytime, in support of high-quality care.
- Provide individuals with secure private lifetime record of key health history and care.

EHR Advantages and Issues

➤ Potential advantages

- Reduce errors
- Improve quality of care
- Improve efficiency

➤ Issues

- Implementation of standards, such as HL7, for interoperability
- Feasibility and cost
- Adoption and practice disruption
- Culture
- Technology: making the wrong choice
- Legal barriers
- Data protection: privacy, security, consent

EHR Purpose

- Provide an electronic equivalent of an individual's health record for use by providers and staff **across more than one health** care organization
- Support efficient, high-quality integrated health care, independent of the place and time of health care delivery

Functions of the EHR (1)

1. Supports structured data collection using a ***defined vocabulary***.
- 2. Accessible at any or all times by authorized individuals.***
3. Contains a *problem list* - patient's clinical problems and current status
4. Supports *systematic measurement and recording of* data to promote precise and routine assessment of the outcomes of patient care
5. States the logical basis for all diagnoses or conclusions as a means of documenting the *clinical rationale for decisions* about the management of the patient's care.

Functions of the EHR (2)

1. Can be *linked with other clinical records of a patient*—from various settings and time periods—to provide a longitudinal (i.e. lifelong) record of events that may have influenced a person's health.
2. Can assist the process of *clinical problem solving* by providing clinicians with decision analysis tools, clinical reminders, prognostic risk assessment and other clinical aids.
3. **Can be *linked to both local and remote databases* of knowledge, literature and bibliography or administrative databases and systems so that such information is readily available to assist practitioners in decision making.**
4. Addresses *patient data confidentiality*.
5. Can help practitioners and health care institutions *manage the quality and costs of care*.

Benefits 1/2

- Staff to physician ratio decreased below national ratio average.
- Practice overhead costs will be reduced.
- Patient perception of practice improved.
- Better patient coverage during off hours since information was more consistent, complete and accessible.

Benefits 2/2

- e-Prescribing improves patient safety (instructions, warnings and legibility)
- Sharing data efficiently outside the practice.
- improves quality of care at other provider organizations
- Rewards (financial) for documenting clinical performance
- Improved quality of life for physicians

Components of EHR

➤ EPR

➤ PHR

➤ EMR

Electronic Patient Record (EPR)

- Primarily used by nurses, pharmacists, and allied health professionals:
- Electronic set of information about a single patient.
- An electronic patient record system is specifically designed to provide patient records electronically.
- Not necessarily restricted to a single clinical information system.

Personal health record (PHR)

➤ **PHR** is a tool **you** can use to gather, organize and manage **your** health care information, including medical history, medications, immunizations, health care providers, etc.

➤ formation?

- ✓ Emergency contacts
- ✓ Current medications and dosages
- ✓ Allergies (to foods, drugs and other substances)
- ✓ Important events, dates, and hereditary conditions in your family history
- ✓ A list and dates of significant illnesses and surgical procedures
- ✓ Results from recent doctor visits
- ✓ Important tests results; eye and dental records, immunization records

Electronic Medical Record (EMR)

- **EMR is primarily** used by physicians and office staff
- A record of periodic care, typically owned by a single provider organization, which captures and manages patient data.
- It is a complete patient record accessible from a single, automated health care provider system.
- Single computer application for recording and viewing data related to patient care

Goals of Using the EMR

- Provide a single, uniform medical record.
- Ability to access medical records from any location.
- Improve documentation and coding.
- Improve research / clinical trials data / enhance quality.
- Reduce transcription and other rising costs.

Benefits of EMR Systems

1. **Improvement** in Legibility of Clinical Notes
2. **Decision Support** for Drug Ordering, Including Allergy Warnings and Drug Incompatibilities
3. **Reminders** to Prescribe Drugs and Administer Vaccines
4. **Warnings** for Abnormal Laboratory Results
5. **Support** for Program Monitoring, Including Reporting Outcomes, Budgets and Supplies
6. **Support** for Clinical Research
7. **Improve Management** of Chronic Diseases (Diabetes, Hypertension and Heart Failure).

Why EMR

- Selection and implementation of an EMR requires attention to detail and an intimate understanding of **practice and Clinical processes**.
 1. What problem(s) the system should solve, or
 2. Which processes it is intended to improve?

Why EMR

- Better understand your **clinical practice style** are often stymied by difficult to pin-point inefficiencies.
- For example,
 - why are charts frequently difficult to find?
 - What accounts for the lag-time between a patient's arrival and placement in an exam room?
 - Where is the report for the chest -x-ray done 3 weeks ago?
- The answer to each of these questions can only be answered by a thoughtful review of the procedures involved.
- However lack of an up-to-date clinical policy and procedure document is a significant part of the underlying problem.

EMR Versus EHR

EMR	EHR
A record of medical care created, managed, and maintained by one health care organization (intra-organizational)	A repository of individual health records that reside in numerous information systems and locations (inter-organizational)
Integration of health care data from a participating collection of systems from one health care organization	Aggregation of health-related information into one record focused around a person's health history, i.e., a comprehensive, longitudinal record
Consulted by authorized clinicians and staff within one health care organization.	Consulted by authorized clinicians and staff across more than one health care organization
Data continuity throughout one health care organization	Data interoperability across different organizations

EHRs Versus Paper Records

- EHRs can make a patient's health information available when and where it is needed
- EHRs can bring a patient's total health information together in one place, and always be current
- EHRs can support better follow-up information for patients
- EHRs can improve patient and provider convenience

EHRs Versus Paper Records

➤ EHRs

- Can link information with patient computers to point to additional resources
- Don't just “contain” or transmit information, they also compute with it
- Can improve safety
- Deliver more information in more directions
 - While reducing “paperwork” time for providers
- Improve privacy and security
- Reduce costs

EHR Effect on Patient Care Safety

- Reduces the need to repeat tests
- Reduces the number of lost reports
- Supports provider decision making

EHR Effect on Efficiency

- Improves accessibility of patient information
- Integrates data from multiple internal and external sources
- Facilitates the co-ordination of health care delivery

EHR Effect on Patient Outcomes

- Has the potential to
 - Improve the quality of patient care
 - Help providers practice better medicine
 - Provide seamless exchange of information among providers

Uses of the Medical Record

- **Main purpose:** Facilitate patient care
- **Historical record:** What happened, what was done
 - Communication among providers (& patients)
 - Preventive care (immunizations, etc)
 - Quality assurance
 - Legal record
- **Financial:** coding, billing
- **Research:** prospective, retrospective

Barriers of EMR

- Expense
- Staff resistance
- Time & effort required
- Incompatibility of hardware/software
- Security

- Lack of demonstration site
- Data/chart conversion
- Increase documentation
- Lack of technical expertise

EMR Implementation

- Different implementation paths for different practice sizes and specialties
- Realistic timeframes
- Staff involvement
- Workflow changes
- Data conversion: scanning
- Support and maintenance
- Backups and recovery

EMR Implementation

➤ Policies & procedures

- Privacy, confidentiality, security
- Medicolegal requirements
- Backups and disaster recovery
- Support and maintenance
- Have flexible timetables
- Appoint a project manager
- Assign responsibilities
- Modify schedules

- Plan and test, plan and test
- Start immediately following training
- Implementation never ends

Main Activities in EMR

- Patient Registration
- Appointments
- E-prescribing
- Lab/Imaging results
- Health History

Types of EMR Systems

1. Free

- Ad-supported
- Free (often do not include the whole software and are missing important features)

2. Open source

3. Paid

Free EMR Systems

- Free EMR system is ideal for small medical practices and most do not require programming knowledge.
- Free EMR companies have different ways of earning revenue:
 - ✓ Selling ads that appear in their system
 - ✓ Selling **anonymized patient data**
 - ✓ Charging patients for access to some applications (e.g., The patient portal) and/or enticing practices to purchase premium expansions
- **The target customer:** small and midsize practices
- **Major pros:** No upfront costs.
- **Major cons:** Free EMR software versions often exclude important features that are critical to a practice's workflow.

Example of Free EMR Systems: 75Health

Functionalities include: E-prescribing, clinical decision support, built-in scheduler, drug list management, vaccination tracking and more.

The screenshot displays the 75Health EMR system interface. The top navigation bar is green with the 75Health logo and a back arrow. On the right, it shows the user 'Dr. Demo Doctor' and links for 'Help', 'Home', 'Settings', and 'Sign out'. A 'Need Help?' link is also present. The left sidebar contains a menu with 'Patient' (selected), 'New', 'List', 'Medical History', 'Appointment', 'Receipt', 'Settings', 'Help', and 'Sign out'. The main content area is titled 'Patient Medical Record'. It includes a 'Provider:' field with a timestamp '2014-12-04 17:56'. Below this is a table with patient information:

Patient KPID	Patient Name	Email	Birth Date	Age
3022-407	Demo Patient	demopatient1@g.com		

Below the table is a 'Patient Visit Reason' text area containing 'Fever and cold'. At the bottom right of the form are buttons for 'New Medical Record' and 'Print'. A dropdown menu is also visible. At the bottom, there is a list of expandable sections: 'Family Health History', 'Social Habits', 'Allergy', 'Vaccine and Immunization', 'Vitals', and 'Symptom'.

2. Open Source EMR Systems

- Open source EMRs are like blueprints that let you build your own electronic records system
- You can download the software source code and customize it to their exact specifications
- Depending on the EMR, you may have to get a **licensing agreement** from the original creator
- While the source code is often free or low-cost, practices may end up having to hire software developers or consultants to help with their EMR's design, implementation and training.

2. Open Source EMR Systems...

- **The target customer:** Providers with access to technical staff, contractors or consultants
- **Major pros:** The open source community is widely known to be a collaborative group that constantly works on improving the software and sharing their results so you can benefit from their additions
- **Major cons:** The technical know-how required to set up and maintain these systems can be intimidating for the average practice.

Example of Open Source EMR Systems: OpenEMR

- **OpenEMR** is considered the most popular open source EMR by many, and with good reason
- It includes practice management functionalities, such as scheduling and billing.

The screenshot displays the OpenEMR web interface for patient management. At the top, there are buttons for 'NEW PATIENT' and 'CLEAR ACTIVE PATIENT'. The current patient is identified as 'Phil Belford (1)' with a DOB of '1972-02-09' and age of '45'. A dropdown menu for 'Encounter History' is also visible.

On the left side, there is a 'Hide Menu' button and a list of navigation options: Users, Addr Book, Practice, Codes, Layouts, Lists, ACL, Files, Backup, Rules, Alerts, Patient Reminders, Other, Reports, and Miscellaneous. At the bottom left, there is a 'Find:' search bar with options for 'Name', 'ID', 'SSN', 'DOB', and 'Any'.

The main area is titled 'Search or Add Patient'. It features a 'Who' tab (checked) and a form for adding a new patient. The form includes fields for Name (with a dropdown set to 'Unassigned'), External ID, DOB, Sex (with a dropdown set to 'Unassigned'), S.S., License/ID, Marital Status (with a dropdown set to 'Unassigned'), User Defined (with four input fields), and Billing Note. Below the form, there are checkboxes for 'Contact', 'Choices', 'Employer', 'Stats', 'Misc', 'Guardian', and 'Insurance'. At the bottom of the form, there are 'Search' and 'Create New Patient' buttons.

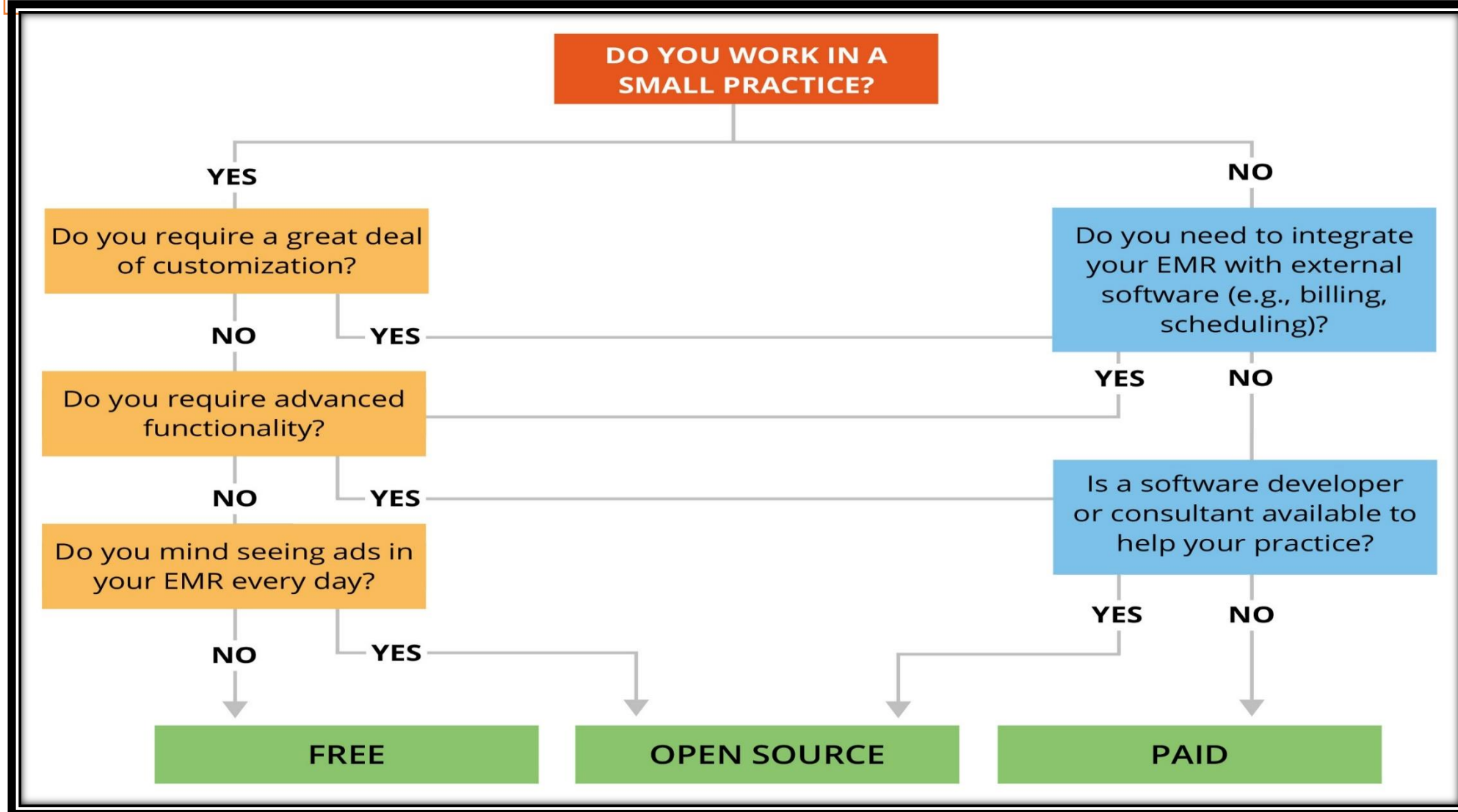
3. Paid EMR Systems

- What you pay for this type of EMR depends on your practice size and preferred deployment model (On-premise or Cloud-based)
- On-premise systems typically require fees for licenses, servers, implementation, training and ongoing technical support
- Web-based systems typically have lower upfront costs, but include monthly fees for licensing, technical support and/or upgrades
- Training and ongoing technical support are usually more accessible if you have paid software (versus open source)
- If something goes wrong, you have someone to call instead of doing it yourself.

3. Paid EMR Systems...

- **The target customer:** Midsize-to-large practices and hospitals
- **Major pros:** More robust functionalities and the ability to easily scale with growing practices
- **Major cons:** If you end up disliking your system, you may be stuck with it for years until you get enough money for a different one.

What type of EMR is right for you?



Chapter-3

Telemedicine/ Telehealth

Chapter objectives:

At the end of this chapter students will be able to:

- Define telemedicine, tele-health, tele-care, Video-conferencing, Tele-monitoring
- Know the types of Telemedicine
- Understand Concepts of telemedicine system
- Identify benefits and limitations of telemedicine.
- Understand technologies used in TM
- Know the application area of TM

Brain storming



1. What is telemedicine, tele-health?

Definitions

- Delivery of health care services, where distance is a critical factor, health professionals use ICT for the exchange of valid information for Diagnosis, Treatment and prevention of disease and injuries, research and evaluation, and for the continuing education of health care providers.
- Is the use of telecommunication and IT in order to provide clinical health care at a distance.
- It improves access to medical services that would often not be consistently available in distant rural communities.

Definitions...

- **Tele-health:** It is the use of telecommunication techniques for the purpose of providing telemedicine, medical education, and health education over a distance.
- **Tele-care:** Tele-care is a term given to offer remote care of elderly and vulnerable people, providing the care and reassurance needed to allow them to remain living in their own homes

Definitions...

- **Tele-monitoring:** The process of using audio, video, and other telecommunications and electronic information processing technologies to monitor the health status of a patient from a distance.
- **Video conferencing:** Real-time, usually two-way transmission of digitized video images between two or more locations.

History of TM

- The exact date when telecommunications first were used in health care is unknown.
- It is known that telegraph was used during American Civil War to transmit casualty lists and order medical supplies.
- By 1900, the telephone was in use, and physicians were among the first to adopt it.
- Telemedicine began over 40 years ago with a small number of hospitals extending care to patients in remote areas.

History of TM...

phases of telemedicine development	Timescale of Development
Telegraphy	1840s- 1920s
Radio	1920s onwards
Television	1950s until 1980s
Digital technologies	1990 onwards (main

Telemedicine Technology – Evolution

- Point to Point System - One patient end connect to One Specialist within the hospital
- Point to Multi Point System - One patient end at a time connect to any of the specialists' end within the hospital.
- Multi Point to Multi Point System - Several patients' end simultaneously connect to different specialists' end at different hospitals at different geographical locations

Progression and Current uses of TM

- The earliest objective of telemedicine was to assist in the **delivery of health care to persons geographically remote** from physicians or a medical center.
- Recently, projects developed to promote **remote consultation, diagnosis and clinical services** via various forms of telecommunication.

Types of Telemedicine

- The scope and categorization of telemedicine practice have changed as the technology has developed
- Currently, we can identify four different types:
 - Tele-consultation
 - Tele-education
 - Tele-monitoring
 - Tele-surgery

Tele-consultation

- The medical consultation is at the heart of clinical practice. Therefore, Teleconsultation is to support clinical decision making is the most common TM service.
- Studies have shown that tele-consultation accounts for about 35% of the usage of telemedicine networks.
- A tele-consultation can take place between two or more careers without patient involvement or between one or more careers and a patient.
- The simplest example is a telephone conversation between two physicians to obtain a second opinion.

Tele-education

- We can distinguish several types of tele-education depending on who is the recipient and what is the purpose of the transmission:
 - Clinical education from teleconsultation
 - Clinical education via the internet
 - Academic study via the internet
 - Public education via the internet

Tele-monitoring

➤ Tele-monitoring is the use of a telecommunications link to gather routine or repeated data on a patient's condition.

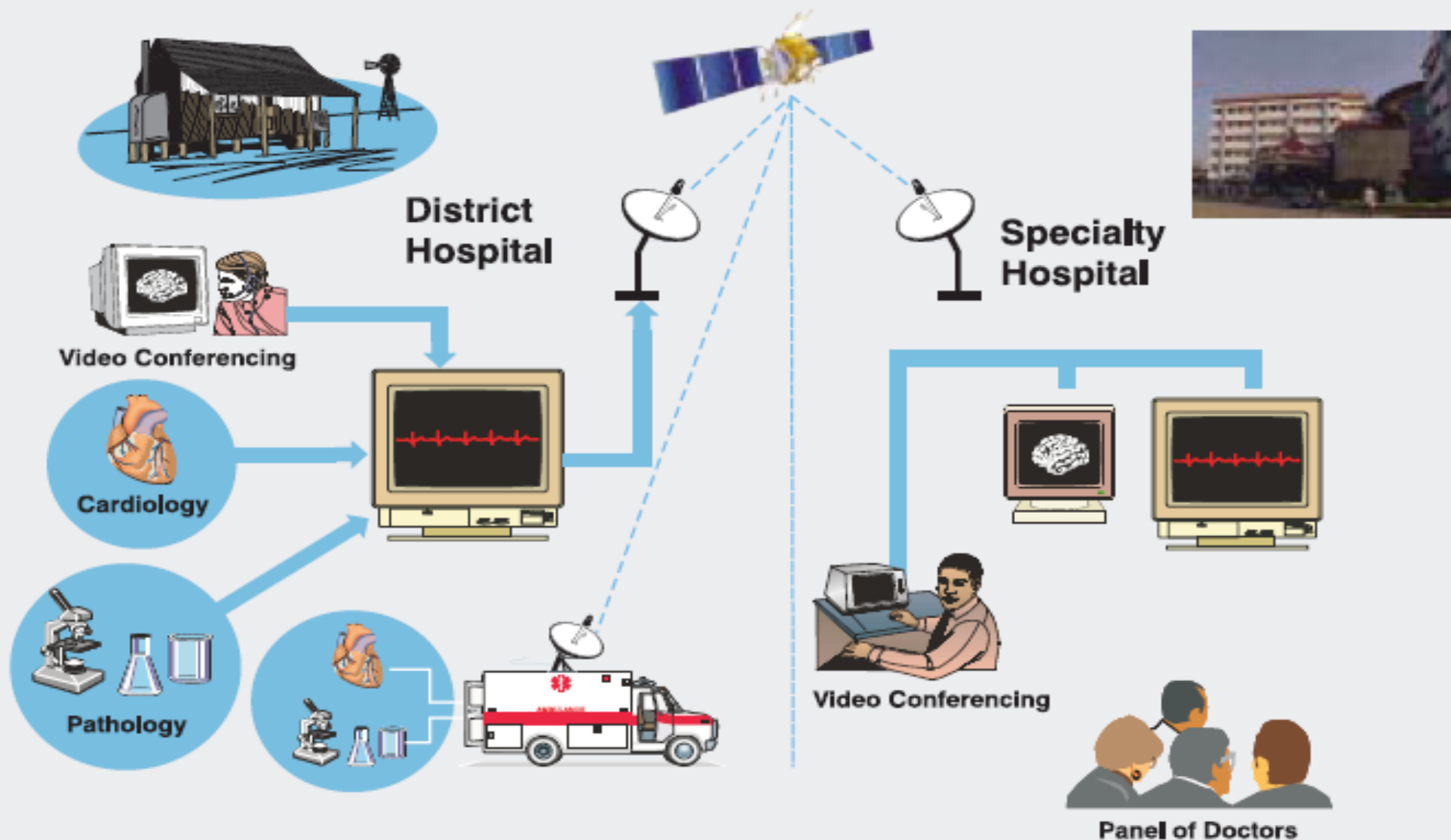
Can be.....

- ✓ **Manual** if the patient records the data and transmits them by telephone, computer/modem system
- ✓ **Automatic** so that continuous data can be submitted either in real time or in store-and-forward mode

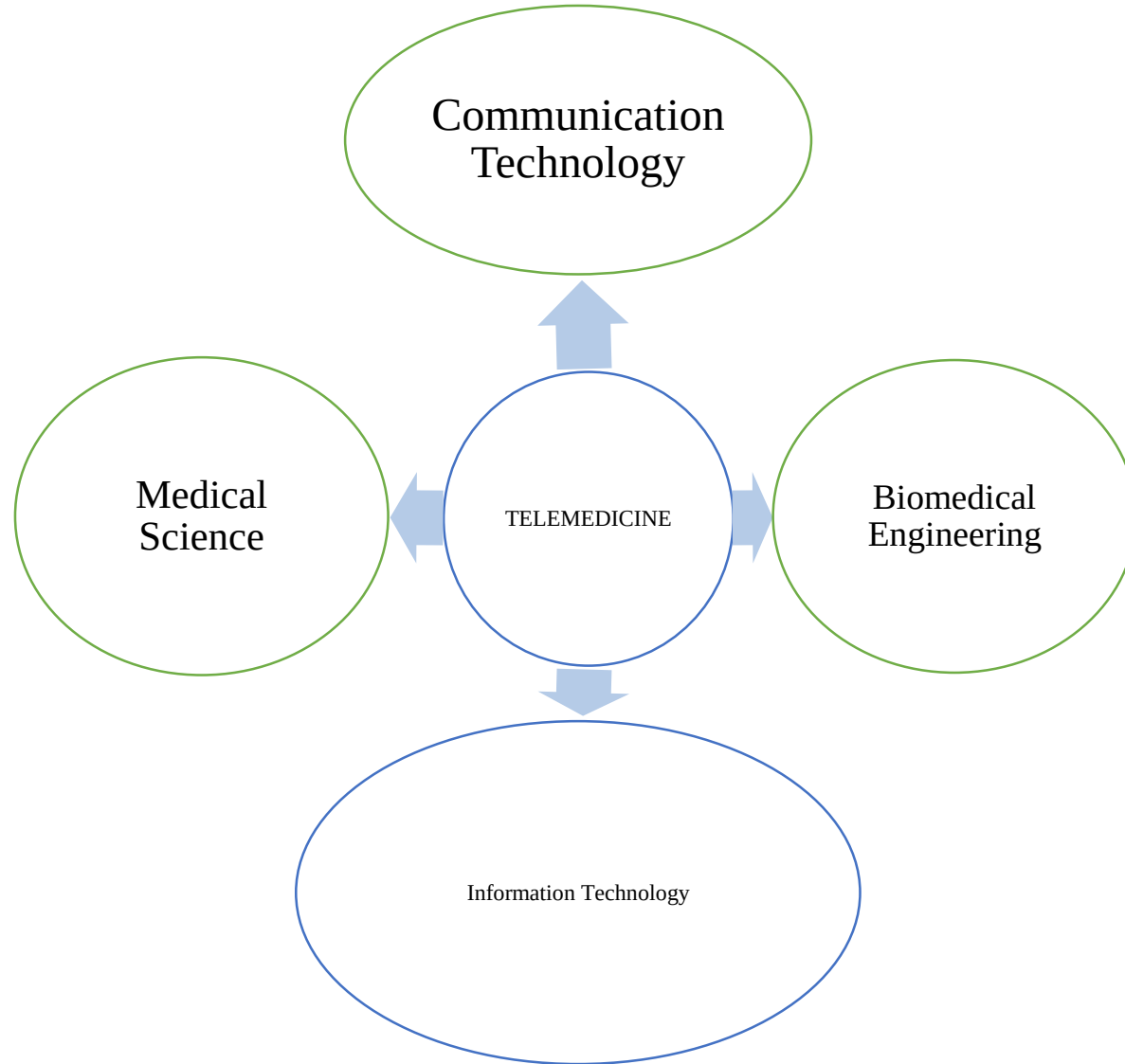
Tele-surgery

- *Tele-surgery is in its infancy stage compared to other 'tele' applications*
- It is practiced in two ways:
 - ✓ **Tele-mentoring:** assistance is given by specialists to surgeons carrying out a surgical procedure at a remote location via video and audio connection that can extend elsewhere in the building or over a satellite link to another country (potential for tele-education)
 - ✓ **Tele-presence surgery** which guides robotic arms to carry out remote surgical procedures

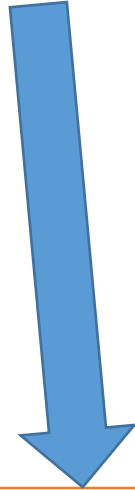
TELEMEDICINE CONCEPT



Components of TM technology



Brainstorming



➤ What are the main benefit of telemedicine?

Benefits of Telemedicine

- Improved Access
 - Covers previously unserved or underserved areas
 - Access to Specialty Clinics
- Improved quality of care
 - Enhanced decision making through collaborative efforts
- Reduced isolation of healthcare professionals
 - Peer and professional contacts for patient consultations and continuing education
- Reduced costs
 - Preventing Patient Travel
 - Traveling a Provider
 - Lower Reimbursement Costs

Medical constraints and challenges

- Need and will of cooperation between medical sites
- Complementary function of involved institutions and organisations
- Acceptance of technology and change of working environment

Potential Problems

- Resistance from the doctor's side
- The potential benefit derived from the introduction of telemedicine services is not brought to the knowledge of doctor's community and healthcare administrators
- Systems must be focused on the needs of the medical profession and patients, and not forced by technology
- Telemedicine may not be seen cost-effective since it often enhances the service rather than driving the process more efficient

Limitations in Developing Countries

➤ **Shortage of general practitioners**

- many rural clinics are run by nurses only

➤ **Shortage of medical specialists**

- many countries have no dermatologists, pathologists, ophthalmologists, radiologists, etc.

➤ **No access to current information**

- medical libraries have no current material, no continuing medical education (CME)

➤ **Limited Communication Infrastructure**

- Existing telephones are often slow and too unreliable for stable internet access

➤ **Limited financial resources**

Limitations in Developing Countries...

Limited Transportation



Telemedicine System Requirement

- Trained people
- Technology
- Funding
- Process

Telemedicine Requirements

- **Hardware:** network devices, computer systems, medical equipment
- **Software:** EMR/EHR
- **Human Resources:** technical users (e.g. physicians, nurses)
- **Telecommunication connectivity:** telephone, internet, email
- **Training and support**

Telemedicine Technologies

The technology required for a telemedicine link can be divided into three categories:

1. Equipment to capture the clinical information at each site
 - High definition Examination Camera
 - Digital medical equipment's (e.g. stethoscopes, vital sign monitors, retina cameras, ultrasound, ECGs)
2. The telecommunications link needed to transmit this information between the sites (e.g. telecommunication network, Internet, LAN, WAN)
3. Equipment to display the information at each site (e.g. computers, projectors, TV sets, mobile, tape recorder, speaker)

Telemedicine Technologies...

- Before the technology can be selected, it is necessary to consider the nature of the information to be transmitted between the sites
- Factors to be considered include:
 1. The types of information to be transmitted,
 2. The quantity of information to be transferred
 3. Security and privacy.

Data/Information exchange in TM

- Not all information will be needed at every site
- Tele-psychiatry application will probably require ordinary commercial videoconferencing equipment instead of very high-quality audio/video signals.
- Tele-monitoring service will require only data and text transfer, without audio and video.
- The clinical need for any telemedicine project must be carefully assessed before making decisions about what equipment will be required to avoid disproportionate setup

Data/Information exchange in TM...

Information Source	Type	Typical File Size
Electronic Stethoscope	Audio	100kByte
ECG Recording	Data	100kByte
Chest X-Ray	Still Image	1MByte
Fetal Ultrasound Recording (30s)	Moving image(Video)	10MByte

Data/Information exchange in TM...

- **Quantity of digital information** to be transferred matters the type of technology to be used.
- One method by which the total volume of information can be reduced is to **compress** it first, and then **decompress** it on reception.
- This is not always an acceptable technique for **medico legal** reasons (i.e. only the original raw signal may be considered acceptable)
- A careful assessment of the above needs will determine the **quantity of information** that must be transmitted between the project sites, and the **time frame** over which it must be sent to achieve the desired clinical goals

Telemedicine Application

Today, telehealth encompasses three distinct models of applications:

1. Real-Time services
2. Store-and-Forward (S&F)
3. Remote Patient Monitoring (RPM)

Real-Time (Synchronous)

- **Live and two-way** interaction between a person and a provider using audiovisual telecommunications technology (videoconference)
- Interactive telemedicine
- Services provide real-time interactions between patient and provider, including phone conversations, online communication and home visits
- Many activities such as history review, physical examination, psychiatric evaluations and ophthalmology assessments can be conducted comparably to those done in traditional face-to-face visits
- In addition, "clinician-interactive" telemedicine services may be less costly than in-person clinical visit

Store-and-Forward telemedicine

- Asynchronous Transmission of recorded health history through an electronic communications system to a practitioner, usually a specialist uses information to evaluate the case outside of a real-time or live interaction.
- Involves acquiring medical data (like Medical images, Bio-signals etc.) and then transmitting this data to a doctor or medical specialist at a convenient time for assessment offline
- It does not require the presence of both parties at the same time. Eg; Tele-Dermatology, Radiology, and Pathology are common specialties that are conducive to asynchronous telemedicine.
- A properly structured medical record preferably in electronic form can be a component of this transfer

Remote Patient Monitoring

- Personal health and medical data collection from an individual in one location via electronic communication technologies, which is transmitted to a provider in a different location for use in care and related support known as self-monitoring or testing.
- Enables medical professionals to monitor a patient remotely using various technological devices. This method is primarily used for managing chronic diseases
- These services can provide comparable health outcomes to traditional in-person patient encounters, supply greater satisfaction to patients, and may be cost-effective

Common Health care application areas

1. Rural Health

- One of the greatest challenges in rural health is assuring that medical expertise is available when it is needed and where it is needed
- This is difficult for remote rural healthcare facilities because they are often unable to attract, afford or retain specialty providers.
- Telemedicine helps solve these issues by allowing access to specialists regardless of location
- This is done using live video conferencing, or real-time medical image sharing/communication portals
- The specialist can examine the patient, review vital signs and patient history, provide assessment, diagnosis and treatment. Usually treatment can be delivered locally.
- Telemedicine also helps rural facilities to train and retain clinicians because telemedicine allows on-the-job experience and remote participation in grand rounds

2. Prison Facilities

- Telemedicine allows prison facilities to deliver high quality care without cost and dangers of transportation or need for physicians in the facility.
- Telemedicine has proven effective for primary care, disease prevention, etc
- The US federal prison system and corrections facilities in many states, as well as prisons outside the US, have implemented telemedicine as a means to deliver routine healthcare

3. School-Based Health Centers

- School-based TM programs significantly keeping children healthier & reducing absences due to illness.
- TM helps manage chronic conditions such as asthma, diabetes & obesity.
- School nurse is an isolated provider. TM allows a school nurse to remotely access expert medical opinion in need.
- This means a student can be treated without missing school for a doctor appointment

4. Disaster Relief

- The benefits for Disaster Relief are similar to rural health.
- Telemedicine helps deliver healthcare quickly after a disaster has hit
- It provides on-site healthcare providers with rapid access to advanced medical expertise and emergency triage capabilities that would not be possible without the use of real-time patient assessment technologies

5. Shipping and Transportation

- The use of telemedicine on cargo ships, cruise ships, private yacht, or commercial airplanes can help avoid the high cost of evacuations and unscheduled diversions that are caused from a medical emergency
- Telemedicine allows access to advance healthcare expertise, triage advice as well as diagnosis and treatment regardless of where the ship or plane may be

6. Industrial Health

- Industrial sites such as mines, drilling platforms or industrial campuses highly depend on the health of their employees to operate
- They must respond to an unpredictable set of health needs to support sometimes hundreds of employees
- Telemedicine avoids the high cost evacuations and assures that the worker receives the best possible treatment and is available to support operations as soon as possible

Potential Telemedicine Application in Developing Countries

- Extension of primary health care delivery
- Consultations with specialists (within the country & abroad)
- Medical emergencies and disaster relief
- Education and training
- Access to specialized databases
- Urban to rural for specialty/unavailable care

Major areas of Telemedicine

- Local doctor consults a specialist to treat a patient
 - Tele-consultation
 - Tele-diagnosis
 - Tele-treatment
- For continuing medical education, training for doctors and paramedics from a higher level hospital/institution
 - Tele-education
 - Tele-training
- Regular monitoring for intensive care and emergency
 - Tele-monitoring
- Support during disaster management
 - Tele-support

Privacy issue in TM

- Healthcare providers participating in telemedicine have the same obligation of responsibility for the privacy and security of patient information as those providing face-to-face care

Privacy Guidelines in TM

1. Audio, video, and all other data transmission shall be secure through the use of encryption (at least on the side of the healthcare professional) that meets recognized standards
2. Individuals in charge of technology should familiarize themselves with the technologies available
3. When the patient and/or provider use a mobile device, special attention should be placed on the relative privacy of information being communicated over such technology

Privacy Guidelines in TM...

4. Providers should ensure that access to any patient contact information stored on any device is adequately restricted. Unauthorized persons shall not be allowed access to sensitive information stored on any device, or use the device to access sensitive applications or network resources
5. Videoconferencing software shall allow only a single session to be opened, although the session may include more than two sites/participants
6. Protected health information and other confidential data shall only be backed up to or stored on secure data storage locations

Privacy Guidelines in TM...

7. Professionals and patients shall discuss how this information will be stored, and how privacy will be protected.
8. Access to the recordings shall only be granted to authorized users and should be streamed to protect from accidental or unauthorized file sharing and/or transfer.

Thank you!!

Wollo University
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Introduction: **m-Health**

By: Mulugeta D. (BSc, MPH/HI)

OBJECTIVES

➤ At the end of this chapter the students will be able to:

- ✓ Define mobile technology/device
- ✓ Explain the special features of mobile device
- ✓ Identify the different types of mobile device
- ✓ Describe issues to be consider while selecting mobile device for any mHealth project
- ✓ Define mHealth
- ✓ Recognize the principles of mHealth
- ✓ Identify the current use or application of mHealth
- ✓ explain the current use or application of mHealth

Mobile Technology



mHealth



Group discussion

- ✓ What is mobile device
- ✓ What are the different types of mobile device
- ✓ What are the basic features of mobile device
- ✓ Mobile phone components
- ✓ Limitations of mobile technology
- ✓ Mobile device manufacturers

Terminology

Mobile technology

- ✍ From the Latin word “mobilis” meaning “to move”

- ✍ Able to move freely or easily

- ✍ Mobile technology is defined an electronic device that allows the user to transport, travel and essentially become mobile

- ✍ known as a handheld device

Common forms of mobile technology

- Mobile phones
- Personal digital assistants (PDAs)
- Tablets
- Patient monitoring devices
- Global positioning system (GPS) devices

Features of mobile technology

- Portability
- Accessibility
- Ease of use
- Applications
- Personalization

Limitations of mobile technology

- Privacy and security
- Damage/loss
- Power/battery
- Health effect (**Gadgets addiction**)

Mobile Phone

✍ A portable, handheld communications device connected to a wireless network that allows users to make voice calls, send text messages and run applications.



Mobile phone cont'd...

- ✍ Is a phone that can make and receive telephone calls over a radio link while moving around a wide geographic area
 - ✍ Also Known as
 - ✍ Cellular phone,
 - ✍ Cell phone,
 - ✍ Hand phone or
 - ✍ Simply a phone
- ✍ Real-time voice communication and basic SMS functions serve as the backbone and the current most common use of mobile phone technology.

Mobile phone cont'd...

- ✍ It does so by connecting to a cellular network provided by a mobile phone operator.
- ✍ The service area of each base station is called a cell.

Mobile network operators in Ethiopia

- A mobile network operator also known as a wireless service provider is a provider of services that owns or controls all the elements necessary to sell and deliver services to an end user.
- Ethio Telecom, previously known as the Ethiopian Telecommunications Corporation (ETC), is an integrated telecommunications services provider in Ethiopia,
 - providing internet and telephone services
 - data communications

Mobile phone components and features

Mobile phones Design

- All mobile phones are designed to
 - Work on cellular networks and
 - Contain a standard set of services that allow phones of different types and in different countries to communicate with each other
- Roaming which permits the same phone to be used in multiple countries
- Providing that the operators of both countries have a roaming agreement

Mobile phone components

- The common components found on all phones are:
- A battery: providing the power source for the phone functions
- An input mechanism : The most common input mechanisms are **keypad** and **touch screens**
- SIM card: All GSM phones use a SIM card to allow an account to be swapped among devices
- An International Mobile Equipment Identity (IMEI) number
- Display: Mobile phones have a display device

Input Mechanisms

Keypad Input

- Some mobile phones typically implement a 12-digit keypad.
- This is very intuitive and effective for numerical input but difficult for entering text.
- This is because each key on a keypad represents three characters.

Input Mechanisms

Soft keyboards

- The most approachable method of entering text on mobile devices is by using a soft keyboard, a keyboard displayed on the screen.
- Users enter data by pressing "keys," just as they would on a physical keyboard.
- This method is very easy to learn and use, but it also takes up screen space.
- It is often used as a backup mechanism when other forms of data input are not effective.

Input Mechanisms

Character Recognition

- Character recognition works by interpreting which letters a user is entering on the screen.
- There is usually a specific area on the screen for users to construct the characters.
- The letters have to be constructed fairly accurately for them to be understood.
- This is usually not a problem for small amounts of data, but can be a challenge for users who need to enter data quickly.

Input Mechanisms

Keyboard Input

- Even with the advances being made in other forms of data input, the keyboard remains the most efficient and easy to use (especially for Western languages such as English).
- Mobile devices have many ways to take advantage of keyboard data entry.

Mobile phone features

- The features of mobile phones are the set of capabilities, services and applications that offer to their users
- Applications like a clock, alarm, calendar, calculator and games
- Most handheld devices can also be equipped with
 - Wi-Fi,
 - Bluetooth,
 - GPS capabilities
 - Camera
 - Media player feature
 - Google map

Types of mobile phones

Mobile phones could be classified based on their features as

➤ Feature/Normal phones



➤ Smartphones



Type of mobile phones cont'd...

A feature phone is a cell phone which

- Offer basic telephony
- Low-end mobile phones (with limited capabilities) and
- Low cost in contrast to a modern smartphone

Smart phones

- Handsets with more advanced computing ability
- High-end mobile phones
- The Android, Blackberry and Apple are smartphones that have great apps and fully stocked app stores

Smart phones

- More advanced mobile phone technologies are enabling the potential for further healthcare delivery
- Smartphone technologies are now in the hands of a large number of physicians and other healthcare workers in low and middle income countries
 - 5.27 Billion globally

Smart phones

- The spread of Smartphone technologies opens up doors for mHealth projects such as
 - Technology based diagnosis support,
 - Remote diagnostics and telemedicine
 - Web browsing,
 - GPS navigation
 - Access to web-based patient information

Mobile Devices

Some of the other major areas that should be given special attention include the following

- Device size and weight.
- Available memory for applications and data.
- Processor speed: This will affect the types of applications that will be able to run on the device.
- Screen characteristics such as size, color depth, indoor and outdoor suitability.

Mobile Devices

Some of the other major areas that should be given special attention include the following

- Mobile operating system support: Can it be upgraded when a new version of the operating system is released?
- Expansion slots for adding peripherals such as more memory, wireless modems, GPS receivers, or digital cameras. Looking into what peripherals are available is also a good idea.
- Battery life

Mobile Devices

Some of the other major areas that should be given special attention include the following

- Integrated features such as digital cameras, keyboards, and Bluetooth.
- Software support, including third-party applications, development tools, mobile browsers, hardware drivers, as well as others that may be applicable to your particular situation.

Mobile device manufacturers

- Mobile phone applications have created a new industry within the world
 - Samsung
 - Huawei
 - Apple
 - LG
 - Nokia
 - Sony
 - Motorola

Reading Assignment

- ❖ Evolution of mobile phone
- ❖ Mobile network generation
- ❖ Different mobile technology

mHealth Program and projects

mHealth

- ✍ A term used for the practice of medicine and public health supported by mobile devices.
- ✍ It is medical and public health practice supported by **mobile devices** such as mobile phones, patient monitoring devices, personal digital assistants (PDAs) and other wireless devices [WHO,2011].

Motivation of mHealth

The motivation behind the development of the mHealth field arises from two factors

- Numerous constraints felt by healthcare systems of developing nations
 - High population growth,
 - A high burden of disease prevalence,
 - Low health care workforce,
 - Large numbers of rural inhabitants,
 - Limited financial resources to support healthcare infrastructure and health information systems.

Motivation of mHealth

- Recent rapid rise in mobile phone penetration in developing countries to large segments of the healthcare workforce as well as the population
- Advancements in their innovative application to address health priorities has evolved into a new field of eHealth, known as mHealth.

Mobile phone penetration

- Global Mobile phone penetration rate drastically increasing over the last decade.
- More than 5.27 billion mobile phone users worldwide in 2020 [International Telecommunication Union]
- Improvements in telecommunications technology infrastructure and reduced costs of mobile handsets

Cont'd...

- Almost half the population of Africa subscribed to mobile services in 2015
- In Ethiopia
 - Around 46.75 million mobile users (41%)
 - GTP II plan to reach 64 million

Principles for mHealth

- **Interoperability** — interoperable with sensors and other mobile/non-mobile devices to share vast amounts of data with other applications, such as electronic health records and existing healthcare systems
- **Integration** — integrated into existing activities and workflows of providers and patients to provide the support needed

Principles for mHealth

- **Intelligence** – Offer problem solving ability to provide real time, solutions based on existing data
- **Socialisation** – Act as a hub by sharing information across a broad community to provide support, coaching, recommendations and other forms of assistance.

Principles for mHealth

- **Outcomes** – Provide a return on investment in terms of cost, access and quality of care based on healthcare objectives.
- **Engagement** – Enable patient involvement and the provision of ubiquitous and instant feedback in order to realise new behaviours and/or sustain desired performance.

mHealth Application

Possible mHealth application area

- Education and awareness
- Helpline
- Diagnostic and treatment support
- Communication and training for healthcare workers
- Disease and epidemic outbreak tracking
- Remote monitoring
- Remote data collection

Definition

- **Programme:** a series of projects whose objectives together contribute to a common overall objective.

E.g. Malaria Control Programme

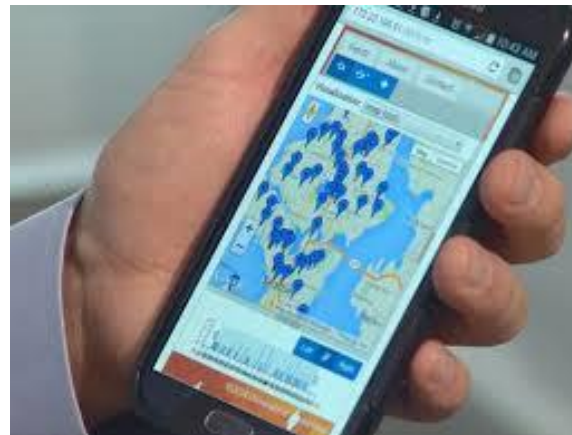
- **Project:** a cluster of activities performed in a **logical sequence** to attain a specific results or purpose in a fixed time frame

E.g. Administration of primaquine in a Malaria endemic area

Program Vs. Project

	Project	Program
Difference	➤ Is specific in objective/purpose. ➤ Has specific area/geographic unit. ➤ Have specific targets groups. ➤ Has clearly determined and allocated fund. ➤ Has specific life.	➤ Has got general objectives. ➤ May not have specific target groups. ➤ May not have clear and detailed financial allocation. ➤ May not have specific time of ending.
Similarities	➤ Has purpose/objectives ➤ Require input (financial, manpower, material, etc) ➤ Generate output (goods and/or services) ➤ Operate over space and time.	

mHealth Application



mHealth Application

Overview

- The mHealth spectrum ranges from
 - Simple mobile phone based applications for the transfer of health information on basic handsets via short message service (SMS) to
 - Highly sophisticated diagnostic applications that rely on advanced equipment and robust back end data systems

mHealth Application Overview

- In Africa, most mHealth interventions are targeting general populations and they have used relatively simple mobile technology and equipment.
- While the field of mHealth is still young, there have been promising early results and lessons from the dozens of pilot programs that have been conducted in a variety of geographic and health system settings.

mHealth Applications

Health call centers

- ✍ Are health care telephone help lines
- ✍ A service created to deliver health care advice services by trained health professionals
- ✍ Address specific health issues such as HIV/AIDS, reproductive health/family planning and drug abuse
- ✍ This approach has been found to overcome widespread health system barriers

Health call centers cont'd...

WHO survey adoption level result

- ✓ The European Region reported more general purpose health call centers (64%) than other regions
- ✓ **African** (17%) and South-East Asia (38%) Regions reported a **low level of pilot stage** activity for health call centers initiatives

Emergency toll free telephone services

- A toll free telephone number or free phone number is a telephone number that is billed for all arriving calls instead of incurring charges to the originating telephone subscriber
- **WHO survey adoption level**
 - The South-East Asia Region reported the highest percentage of emergency toll-free telephone services (88%)
 - The **African Region** reported the **least activity** (31%)

Emergency toll free telephone services

Examples in Africa

- Zimbabwe reported launching a telephone line for citizens to report cholera cases
- Madagascar had a service to support victims of domestic violence
- Togo gave an account of an emergency toll-free telephone service set up for citizens to obtain information on TB

Mobile diagnosis and treatment compliance

Mobile diagnosis

- Using camera-enabled phones
 - **Example: Teleradiology**
-
- Treatment compliance: described as the degree to which a patient correctly follows medical care/advice
 - Worldwide, non-compliance is a major obstacle to the effective delivery of health care

Mobile diagnosis and treatment compliance

- Only about 50% of patients with chronic diseases living in developed countries follow treatment recommendations
- Major barriers to compliance are thought to include the
 - Complexity of modern medication regimens and
 - Lack of comprehension of treatment benefits,
 - The occurrence of un discussed side effects and poor communication
- Efforts to improve compliance have been aimed at providing effective medication reminders and improving patient education
 - Examples: Sending reminder messages by voice or SMS

Appointment reminders

- Appointment reminders are voice or SMS messages sent to patients to schedule or attend an appointment
- Include service follow ups (ANC and immunization) and treatment follow ups (ART and TB)

Examples of applications:

- Short Message Service (SMS) alerts,
- E-mail reminders and

Examples of service reminders in Ethiopia

ENAT Messenger for Maternal Health (SMS focused to remind HEWs)

- Each week, HEWs sent ANC data from their health post to the nearest health center.
- The health center entered the data (sent from the health posts and the ANC data for pregnant women following ANC at the health center level) into the Enat system manually.
- Based on the data entered, the system automatically calculated the expected delivery date (EDD) and sent text message reminders to the respective HEWs one month and one week before a pregnant woman's EDD.

ENAT Messenger cont'd...

- The reminder served as a prompt for the HEW to visit the mother to assess her status.
- After each visit, the HEW are expected to send an SMS confirmation back to the system to confirm the visit.
- The responses were monitored by health care workers at the health center.
- The system could also be used to prompt transport of a mother in labor to a health facility

Awareness and Education

- This series of mHealth strategies focuses largely on the client to improve people's knowledge, modify their attitudes, and change their behavior.
- Needs targeted, timely health education and actionable health information

Examples of applications

- Short Message Service (SMS)
- Multimedia Messaging Service (MMS)
- Video clips
- Images

Awareness and Education

WHO survey adoption level result

- Awareness raising initiatives showed relatively **low levels** of uptake across WHO regions
- The Eastern Mediterranean (28%), European (28%) and Americas (25%) Regions reported using awareness raising initiative

Mobile telemedicine

- The communication or consultation between health professionals about patients using
 - Voice, text, data, imaging or
 - Video functions of a mobile device
- To consult with more highly-trained medical professionals in cases of emergency or urgent treatment

Mobile telemedicine

- As mobile networks develop and camera-equipped mobile devices become more common
 - Telemedicine applications that use photo and video features may allow for more sophisticated consultation

Examples:

- Using a mobile phone camera for tele-consultation and diagnosis of soft tissue injuries in Taiwan, China.
- They found an 85% accuracy rate in the diagnosis,

Adoption level

- The Americas (75%), European (64%) and South-East Asia (62%) Regions reported high rates of adoption of mobile telemedicine initiatives, though a large proportion of these initiatives were in **the pilot phase**.

Decision Support system

- Point of care decision support tools through mobile phones can help ensure quality of care.
 - Information, Protocols, Algorithms, Checklists
 - Rule-based instructions
- **Examples of applications**
 - Stored information “apps”
 - Interactive Voice Response (IVR)

Health surveys

- Health surveys, in the context of mHealth, are defined as the use of mobile devices for health related data collection and reporting
 - Mobile-based data collection software
 - Reporting system : Mobile or web interfaces
- Data exchange : To collect and transmit critical indicators related to health status, health outcomes and system performance for use at all levels of the health system

Health surveillance

- Mobile surveillance is defined as the use of mobile devices to input and transmit data that will be used to track diseases for surveillance program
 - Example: Measles investigation and TB active case finding in vulnerable groups
- Examples of applications:
 - Short Message Service (SMS)
 - E-mail
 - Voice communication
 - IVR
 - GPS also allows real-time mapping of disease outbreaks

Adoption level

- The use of mobile devices for health surveys was **low across all WHO regions**,
- **mHealth surveillance** activity is more prevalent in countries in the low-income (40%) and **lower-middle income** groups (27%) than those in the higher-income groups?????

Mobile referrals

- Notification of incoming referral so that care is immediate
- Improving the referral system
 - Skilled delivery especially for high-risk pregnancies
- Such communication could facilitate transportation via ambulance or other means,
 - While allowing facilities to ensure they have the staff and equipment available to accommodate incoming patients.
 - This is critical to reducing the “third delay” of care

Mobile referrals

- The time sensitive nature of referrals, especially in emergencies, means that they need to be relatively simple (i.e., not require great technological complexity), easy-to-use and capable of enabling rapid action.
- Voice-based systems would allow for real-time communication with facilities to ensure readiness for incoming patients or additional consultation
- **Example:** In emergency obstetric services

Remote monitoring

- In situations where the patient and provider are far away
- Linking mobile phones to a connected, but independent, external device can facilitate remote monitoring of clients
- Extending the reach of health facilities into the community and into clients' homes to
 - Conduct, store, transmit and evaluate diagnostic tests through mobile phones
 - E.g. Blood glucose tests and electrocardiograms (ECGs)
- Examples of applications: Accessory sensors

mHealth Application

Possible mHealth application area

- Education and awareness
- Helpline
- Diagnostic and treatment support
- Communication and training for healthcare workers
- Disease and epidemic outbreak tracking
- Remote monitoring
- Remote data collection

mHealth Application

Education and awareness

- Education and awareness programs within the mHealth field are largely about the spreading of mass information from source to recipient through short message services (SMS).
- SMS messages are sent directly to users' phones to offer information about various subjects, including testing and treatment methods, availability of health services and disease management.
- SMSs provide an avenue to reach far-reaching areas such as rural areas.

mHealth Application

Helpline

- Helpline typically consists of a specific phone number that any individual is able to call to gain access to a range of medical services.
- These include phone consultations, counseling, service complaints and information on facilities, drugs, equipment.

mHealth Application

Diagnostic and Communication

- Diagnostic and treatment support systems are typically designed to provide healthcare workers in remote areas advice about diagnosis and treatment of patients.



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mHealth Application

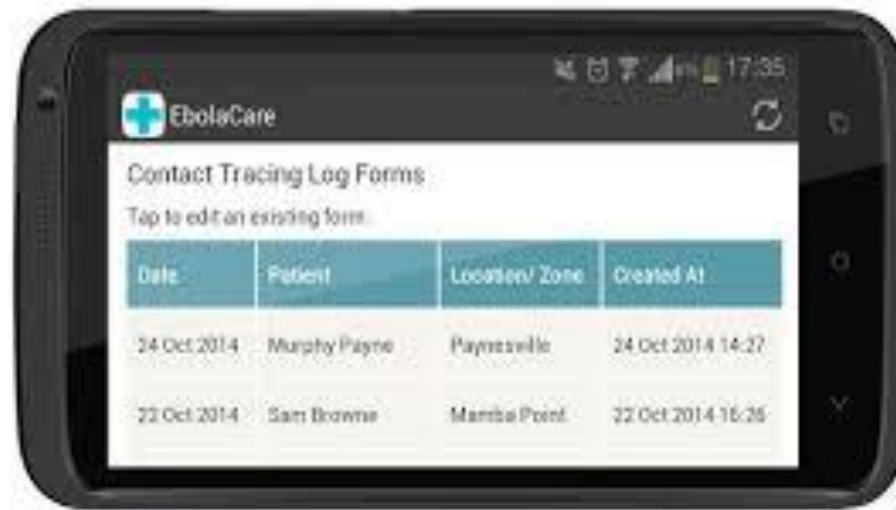
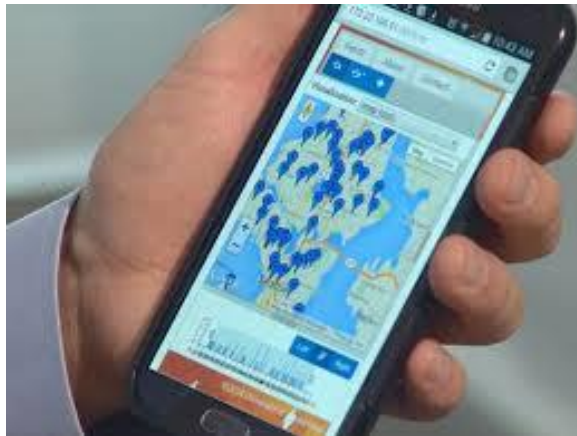
Diagnostic and Communication

- Both diagnosis and treatment support projects attempt to alleviate the cost and time of travel for patients located in remote areas.
- Improved communication projects attempt to increase knowledge transfer amongst healthcare workers and improve patient outcomes through such programs as patient referral processes.

mHealth Application

Disease surveillance and epidemic outbreak tracking

- Mobile phones ability to collect and transmit data quickly, cheaply and relatively efficiently.



mHealth Application

Disease surveillance and epidemic outbreak tracking

- Data concerning the location and levels of specific diseases (such as malaria, HIV / AIDS, TB, Avian Flu) can help medical systems or ministries of health or other organizations identify outbreaks and better target medical resources to areas of greatest need.
- Policymakers and health providers at the national, district, and community level need accurate data

mHealth Application

Treatment support and medication compliance

- Remote monitoring and treatment support allows for greater involvement in the continued care of patients.
- Within environments of limited resources and beds and subsequently a 'outpatient' culture
 - Remote monitoring allows healthcare workers to better track patient conditions, medication regimen adherence, and follow-up scheduling.
 - Through either one- or two-way communications systems.

Challenges of mHealth

- Expense (mHealth infrastructure)
- Stakeholders resistance
- e-literacy
- Time & effort required
- Incompatibility of hardware/software
- Security
- Lack of technical expertise

mHealth in the globe

- ✍ Higher-income countries show more mHealth activity than lower-income countries.
- ✍ Countries in the European region are currently the most active and those in the African region the least active.
- ✍ The capabilities of mobile phones in low and middle income countries has not reached the sophistication of those in high income countries.

mHealth in the globe

- ✍ In US and EU many patients and consumers use their cell phones and tablets to access health information and look for healthcare services.
- ✍ Doctors, nurses and clinicians use mobile devices to access patient information and other databases and resources.
- ✍ In parallel the number of mHealth applications grew significantly the last years.

mHealth in developing countries

- Healthcare access to all reaches of society is generally low in these countries
 - physical distance between doctors and patients,
 - Too few skilled health care professionals and
 - The extraordinary complexity between insular medical systems and costs of health care equipment and infrastructure

mHealth in developing countries

- Largely emerges as a means of providing greater access to larger segments of a population in developing countries, as well as improving the capacity of health systems in such countries to provide quality healthcare
- Governments are expressing interest in mHealth as a complementary strategy for strengthening health systems
- In Ethiopia, got emphasis under HSTP

Issues to consider prior mHealth implementation

- Readiness assessment
- Availability of hardware and software
- Consider sustainability
- Security issue
- Backup
- Select of software and hardware

Reading Assignment

- ✓ History of mHealth
- ✓ mHealth in Ethiopia

Thank YOU!!

Chapter Five!

eHealth architecture and standards

Outline

- Concepts of eHealth architecture
- Standards and interoperability
- Type of standards
- Data exchange standards
- Type of Interoperability
- Terminology Management System

Concepts of eHealth architecture

- eHealth systems that use information and communication technologies for the delivery of healthcare services and information hold great promise for improving global access to healthcare services and health information, particularly in the developing world.
- Delivering on this promise requires more universal eHealth interoperability standards, overcoming technical infrastructure barriers, and addressing privacy, security, and other legal requirements.

Interoperability

- Interoperability is the ability of different information technology systems and software applications to communicate, to exchange data accurately, effectively, and consistently and to use the information that has been exchanged.
- Making Healthcare information systems interoperable will **reduce cost of Health care** and will contribute to **more effective and efficient** patient care.

Interoperability in eHealth System

- Being able to share lifelong EHRs of patients among different healthcare providers; providing clinical decision support through the use of clinical guidelines which require the interoperability of Electronic Health Record Systems (EHRs), laboratory information system (LISs) as part of EHRs and wireless medical sensor devices.

Interoperability

- Interoperability with regard to a specific task is said to exist between two applications when one application can accept data from the other and **perform the task in an appropriate** and satisfactory manner without the need for extra operator intervention.
- Interoperability is the only sustainable way to help partners using **eHealth systems** that was developed and implemented from different vendors, to collaborate harmoniously to deliver quality healthcare.

Types of Interoperability

- **•Technical Interoperability**
 - Neutralizes effects of distance - e.g., hardware, software, networks
- **•Semantic interoperability**
 - Communicates meaning - e.g., data content terminologies (e.g., ICD-10, SNOMED, LOINC)
- **•Process interoperability**
 - Coordinates work processes - e.g., workflow, user roles

NOTE: These three types of interoperability are all required for consistent and timely exchange of health information among partners

- Interoperability of EHRs is confirmed as the fundamental prerequisite for individual health care improvement.
- The causes of eHealth interoperability challenges:
 - Lack of an architectural guide/framework towards development of eHRs.
 - Lack of standard procedures to guide the lifecycle of eHRs.
 - Poor collaboration among private companies or vendors in eHRs development.
 - Different data structures among the developed eHRs.

Behind interoperability there is often a standard.

- Standard is **an agreed way of doing something**. It could be about making a product, managing a process, delivering a service or supplying materials

Standard



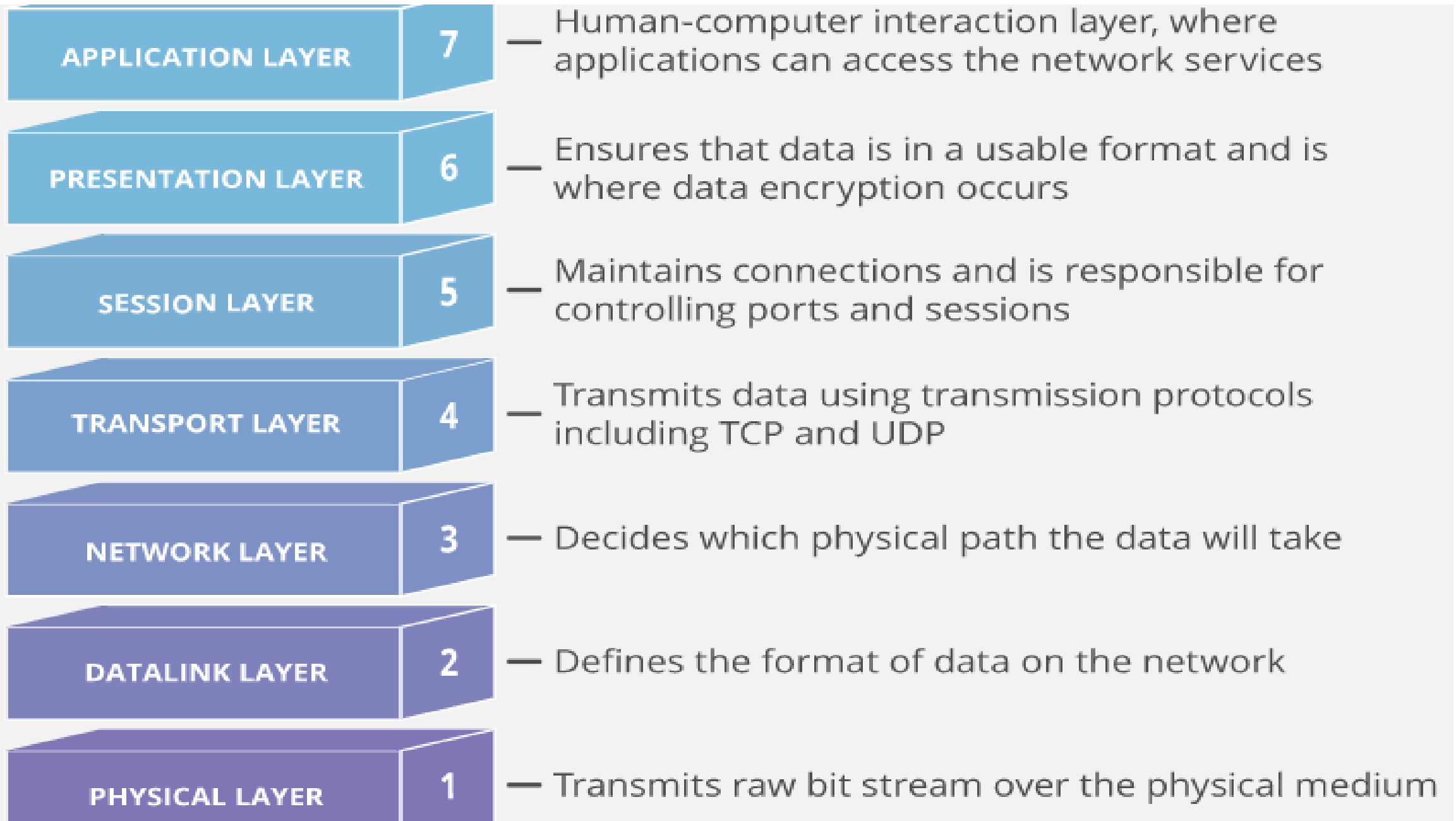
Standards are an everyday thing!

General Standard

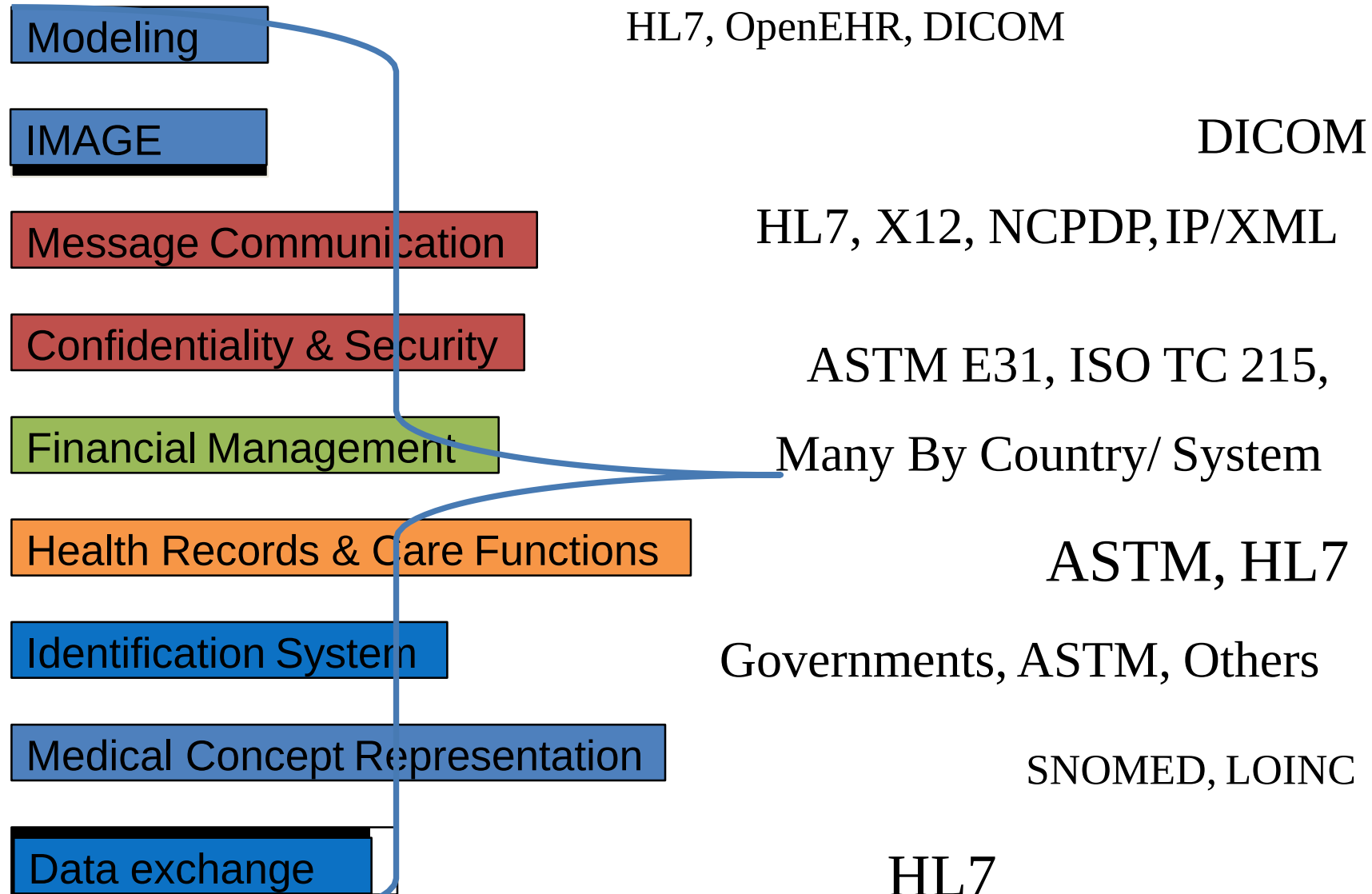
- ✓ CDs,DVDs
- ✓ Bread Size - To Fit Toasters
- ✓ ATM machines
- ✓ Sizes of Shoes,Clothes,
- ✓ Size of Paper A4,A3
- ✓ Etc....

Open Systems Interconnection (OSI) model

- A conceptual model created by the International Organization for Standardization which enables diverse communication systems to communicate using standard protocols.
- OSI provides a standard for **different computer systems** to be able to communicate with each other.
- The OSI model can be seen as **a universal language for computer networking**. It's based on the concept of splitting up a communication system into seven abstract layers, each one stacked upon the last.



Standards Areas for Health



Categories of Standards

- **Identifiers** for providers patients payers facilities, etc. (so that individuals can be recognized across organizations)
- **Medical information** – medications allergies problems etc.
- **Biomedical knowledge** – literature human expertise, guidelines, etc.
- **Messaging between systems**
- **Security and access control**

Standards in eHealth systems

- One of the challenges and opportunities of e-health standardization is the proliferation of multitudinous e-health standards developed by numerous standards-setting institutions.
- One historical challenge is that many of these e-health standards are not interoperable with each other or directly coordinated with each other at an institutional level.
- e-health standardization occurs at all layers of standardization – the physical layer, the data link layer, the network layer, the transport layer, the session layer, the presentation layer, and the application layer there are also technically specific efforts at these various levels.

e-health standardization cont.

- **CEN/TC 251 – Health Informatics:**

- ✓ Is the health informatics technical committee of Comité Européen Normalisation.
- ✓ Primarily publishes standards that address **application** and **content layer** issues in e-health.
- ✓ The standards address aspects of information representation, message standards, electronic health records, and some areas of communication specifications between medical devices.

Continua Health Alliance :

- The Alliance was formed in recognition of the transformation of healthcare from primarily institutional environments to patient homes and other more distributed health delivery environments.
- Some of the technology and healthcare companies included among founding members of this group include Cisco Systems, GE Healthcare, IBM, Intel, Kaiser Permanente, Motorola, and others.
 - ❖ This organization focuses on three areas of e-health:
 1. technologies geared toward health and wellness, particularly for managing weight and preventing the diseases associated with obesity.
 2. chronic disease management and particularly health monitoring and diagnostic systems.
 3. area addresses the aging world population and devices geared toward assisted living for the elderly.

▪ **DICOM Standards for Medical Images:**

- DICOM, short for Digital Imaging and Communications in Medicine, is a set of specifications dedicated to the [standardization of medical images](#).
- DICOM standards specify image file formats, storage protocols, and the processing and transmission of medical images.
- Some of the types of medical imaging standards encompassed by DICOM include: network protocols standardizing communications among devices adhering to DICOM standards; syntactical and semantic information necessary for exchanging medical images; media storage specifications; and file formats for medical images
- The DICOM standards are widely adopted in equipment and information systems used in hospitals, imaging centers, and in providers' offices to produce, display, store, or exchange medical images.

PACS

Picture Archiving & Communication System-(PACS):

Comprises data storage devices, image display devices, database management software and links to image/or image data acquisition devices connected by computer networks.

The major components of PACS

- **Types of Image and data acquisition gateways to the PACS server and archive:**
 1. The data base gate way for *textual data* and
 2. The image acquisition gateway for *imaging data*.
- **A PACS server and archive:** Their major functions are:
 - Receipt of images via acquisition gateways.
 - Extraction of text information via data base gate way .
 - Update the database management system.
 - Determine the destination workstations to which newly generated exams are to be forwarded.
 - Retrieve necessary old patient images.
 - Interface PACS application servers.

■ **HL7 – Electronic Health Information Systems:**

- **A standards organization specifically devoted to the practice of developing standards related to the exchange, storage, and use of electronic health information such as clinical data and administrative information.**
- **HL7 typically refers to both the standards organization and the HL7 family of standards.**
- **The scope of HL7's standards activities is quite large.**
- **Some of HL7's many Work Groups include:**
 - ☐ **Electronic Health Record**
 - ☐ **Clinical Decision Support**
 - ☐ **Security**
 - ☐ **Structured Documents**
 - ☐ **Regulated Clinical Research Information Management**
 - ☐ **Clinical Interoperability Council**
 - ☐ **Public Health and Emergency Response**

- HL7's stated vision is “to create the best and most widely used standards in healthcare” .

ISO/TC 215 – Electronic Health Records

- The International Organization for Standardization (ISO) establishes e-health standards through its Technical Committee, “Health Informatics”.
- Part of ISO’s activity in e-health involves the rebranding of specifications developed in other standards-setting institutions such as HL7 or DICOM.
- For example, ISO 12052:2006 is “Digital Imaging and Communication in Medicine (DICOM)”.

ISO/IEEE 11073 – Medical Device Communication Standards

- Medical device interoperability and communications are critical components of remote care and citizen centric electronic health environments.
- Medical devices addressed under this standard include primarily personal, or end user, health devices such as blood glucose monitors, blood pressure monitors, thermometers, independent living activity hub, weighing scale, pulse oximeters, etc.

ITU-T Multimedia Framework for E-health Applications

- Develop a multimedia framework for e-health applications (and telemedicine in particular).
- Develop a roadmap for e-health standards.
- Construct a general architecture for e-health applications.
- Identify particular characteristics and requirements for e-health applications including video and still picture coding, audio coding.
- Security, and directory architecture.

Terminology management

- A set of activities that ensures correct terms are used consistently in all materials.
- **Terminology** data is stored in a **terminology** database, or term base, and a terminologist is the person **managing** it.

Top characteristics of terminology management systems

- **Consistency:** one-product one name across different units.
- **Centralization:** share the resources of a single, central terminology base.
- **Automation:** extract, search, index, group, and categorize terminology
- **Metadata:** Terms as such that are effectively useless without context.
- **Future-Proofing:** Today's terminology management systems can grow with the content produced for your different needs.

THANK You!!